

ABCD ADHD menarche analysis: Output for longitudinal models

Model 1 longitudinal output

```
Estimate Est.Error  Q2.5 Q97.5
R2      0.736      0.00758 0.721  0.75
```

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.885	0.021	0.844	0.926	1.000	2587	4226

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.089	0.037	0.014	0.165	1.001	955	972

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.142	0.040	0.063	0.220	1.000	4116
menarche_status_p_wave_2Y	0.104	0.041	0.024	0.185	1.001	3983
age_wave_2_sc	-0.119	0.040	-0.198	-0.039	1.000	3731
ethnraceblack_nh	-0.081	0.066	-0.213	0.046	1.001	2923
ethnracehispanic	-0.042	0.058	-0.154	0.073	1.000	2706
ethnraceother	0.004	0.063	-0.121	0.128	1.000	3229

inr_wave_0_sc	0.104	0.042	0.022	0.187	1.001	3273
adhd_traits_wave_0_sc	1.329	0.034	1.263	1.394	1.000	2526
	Tail_ESS					
Intercept	5038					
menarche_status_p_wave_2Y	5419					
age_wave_2_sc	5202					
ethnraceblack_nh	4941					
ethnracehispanic	4964					
ethnraceother	4717					
inr_wave_0_sc	4714					
adhd_traits_wave_0_sc	4355					

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

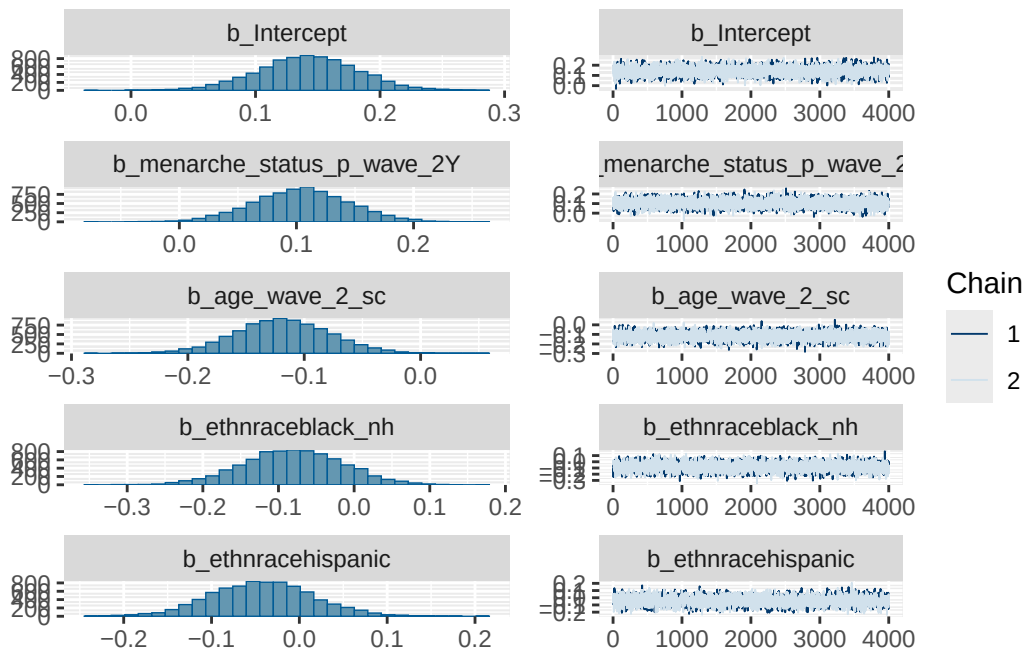
Model 1 longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

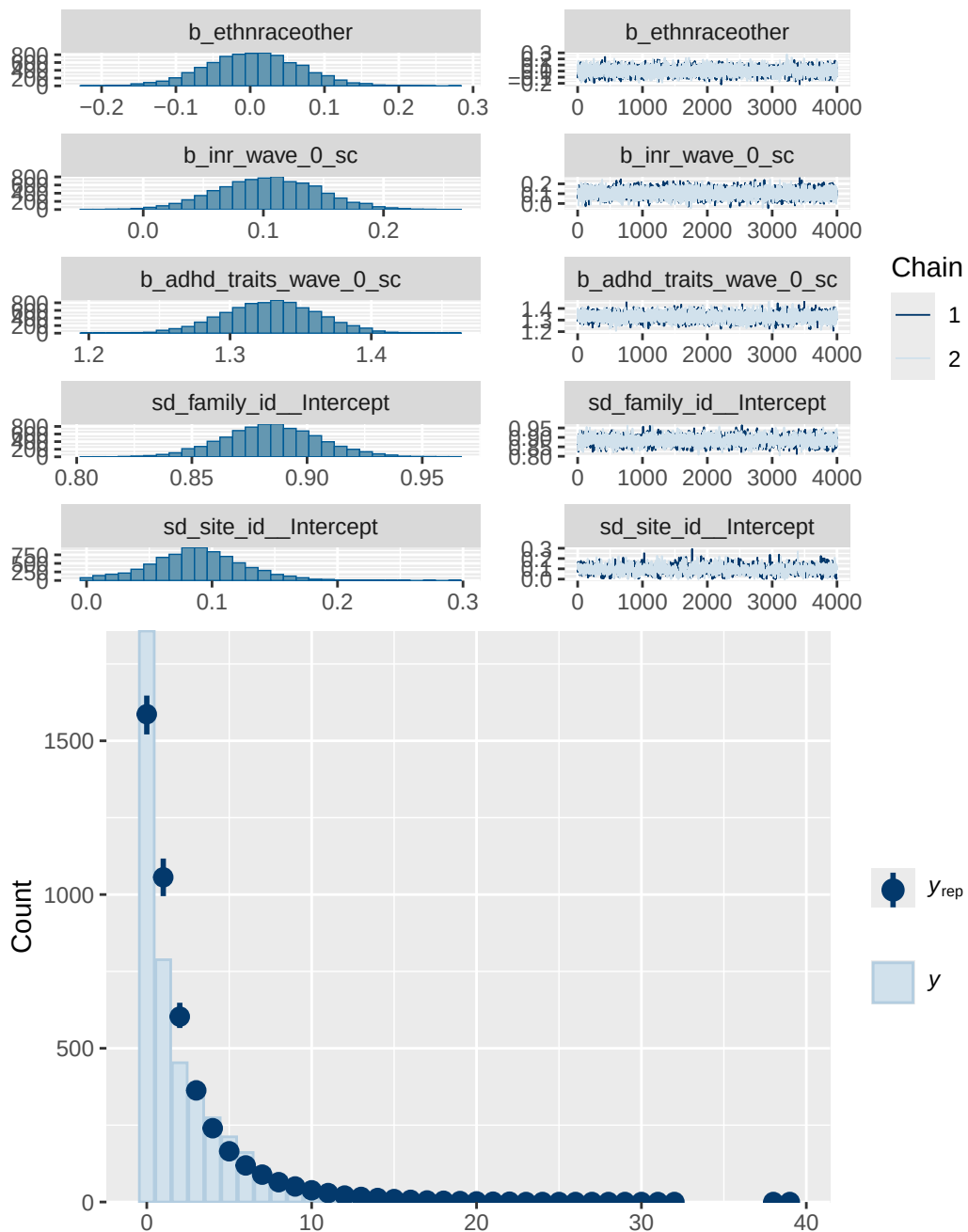
```
# A tibble: 1 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>          <dbl>         <dbl>         <dbl>
1 Menarche Y (Wave 2)    11.0           2.41          20.3
```

Model 1 longitudinal diagnostics

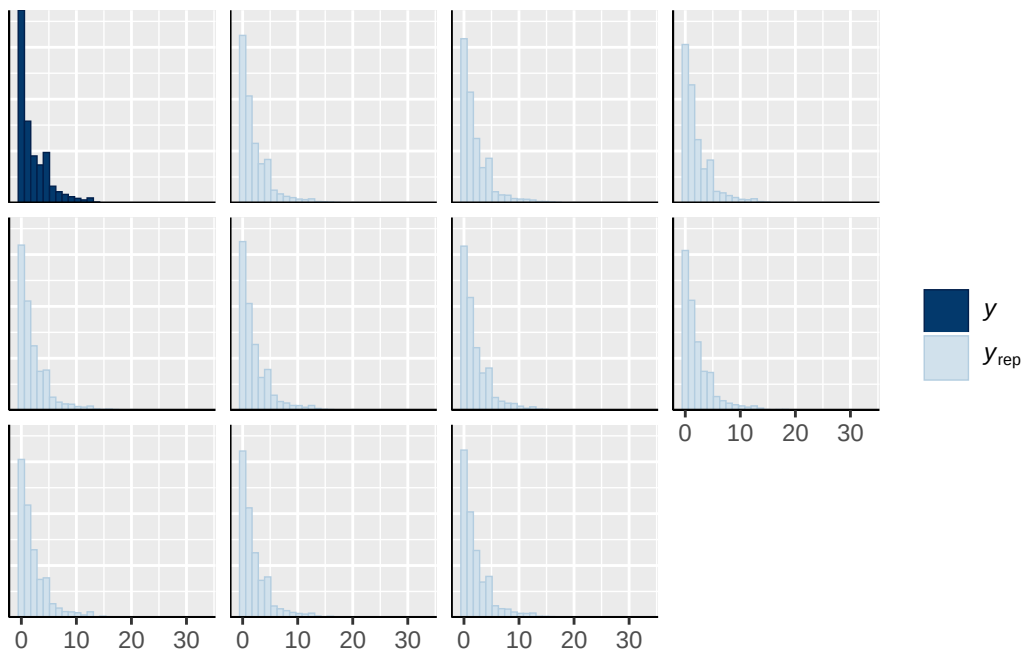
	prior	class	coef	group	resp	dpar
normal(0, 1)		b				
normal(0, 1)		b	adhd_traits_wave_0_sc			
normal(0, 1)		b	age_wave_2_sc			
normal(0, 1)		b	ethnraceblack_nh			
normal(0, 1)		b	ethnracehispanic			
normal(0, 1)		b	ethnraceother			
normal(0, 1)		b	inr_wave_0_sc			
normal(0, 1)		b	menarche_status_p_wave_2Y			
student_t(3, 0, 2.7)	Intercept					
student_t(3, 0, 2.7)	sd					
student_t(3, 0, 2.7)	sd			family_id		
student_t(3, 0, 2.7)	sd		Intercept	family_id		
student_t(3, 0, 2.7)	sd			site_id		
student_t(3, 0, 2.7)	sd		Intercept	site_id		

nlpar	lb	ub	tag	source	user
				(vectorized)	
				(vectorized)	
				(vectorized)	
				(vectorized)	
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				(vectorized)	
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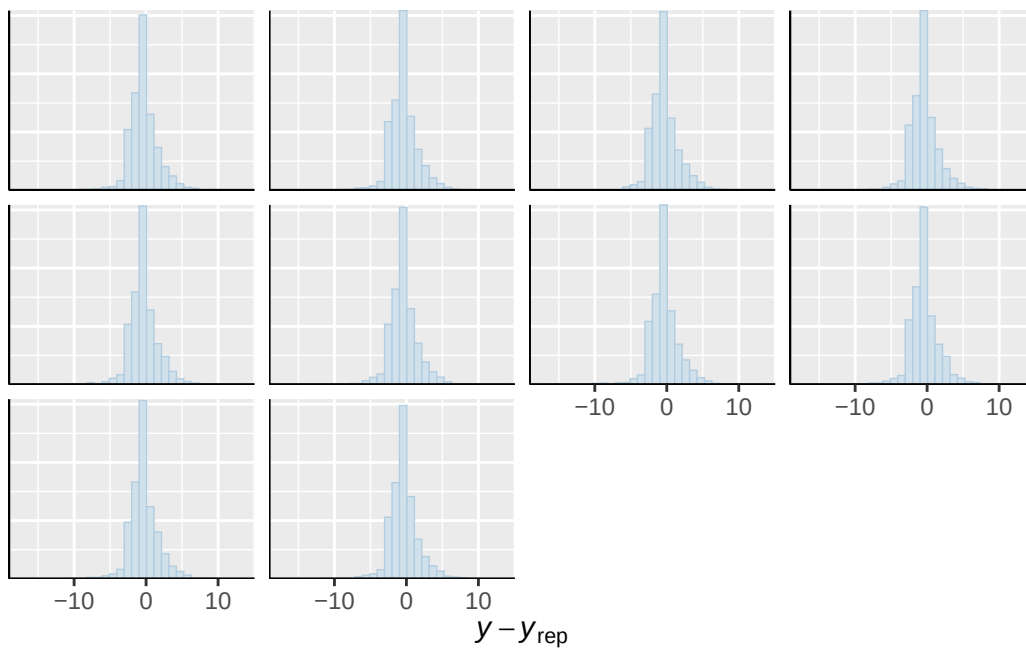




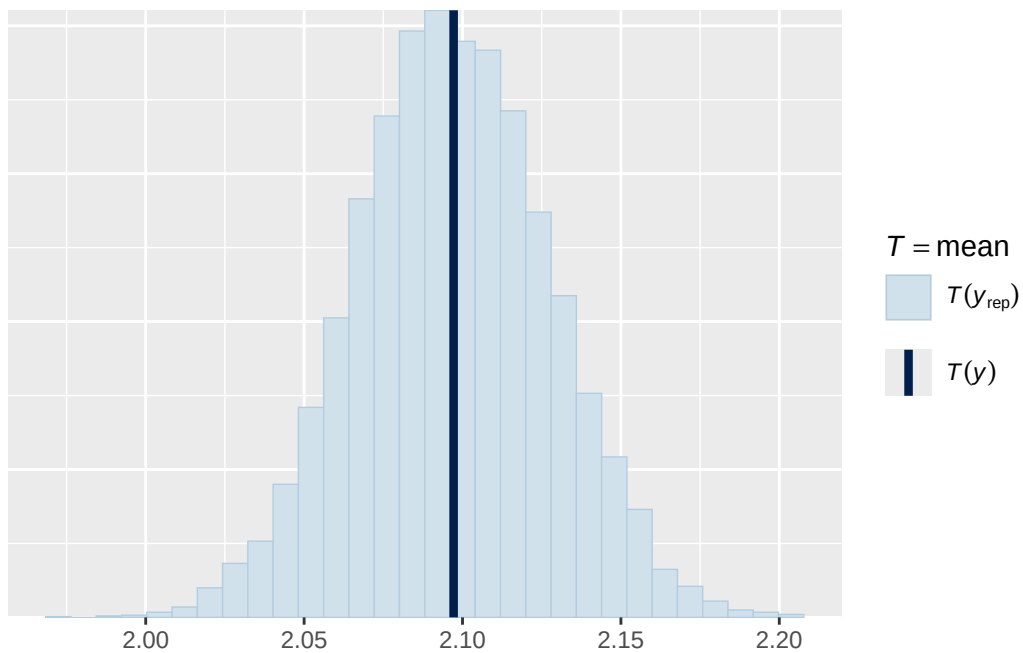
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



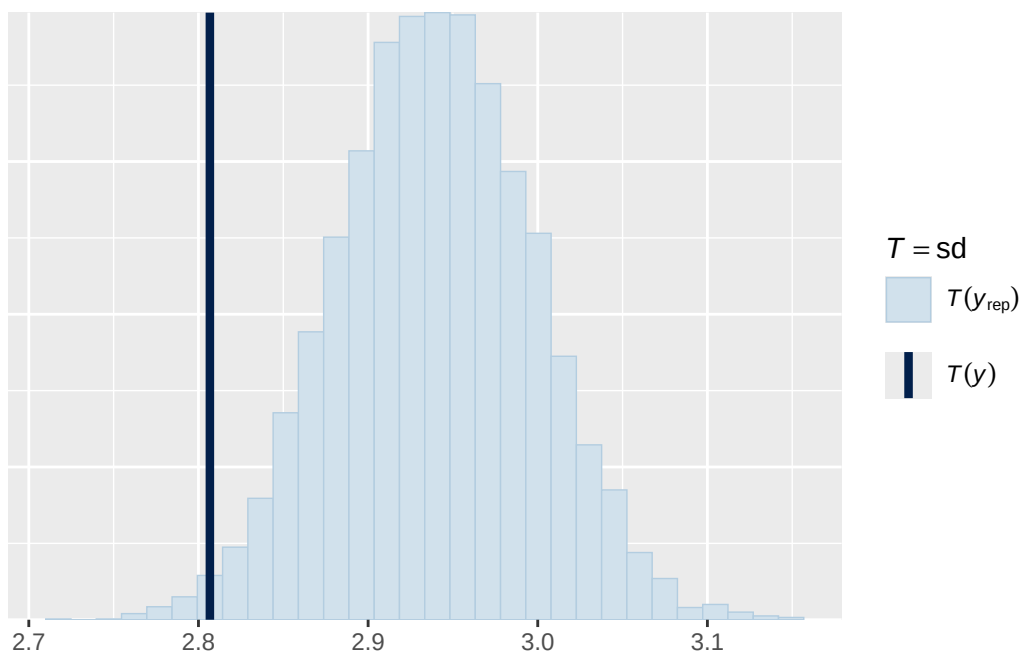
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

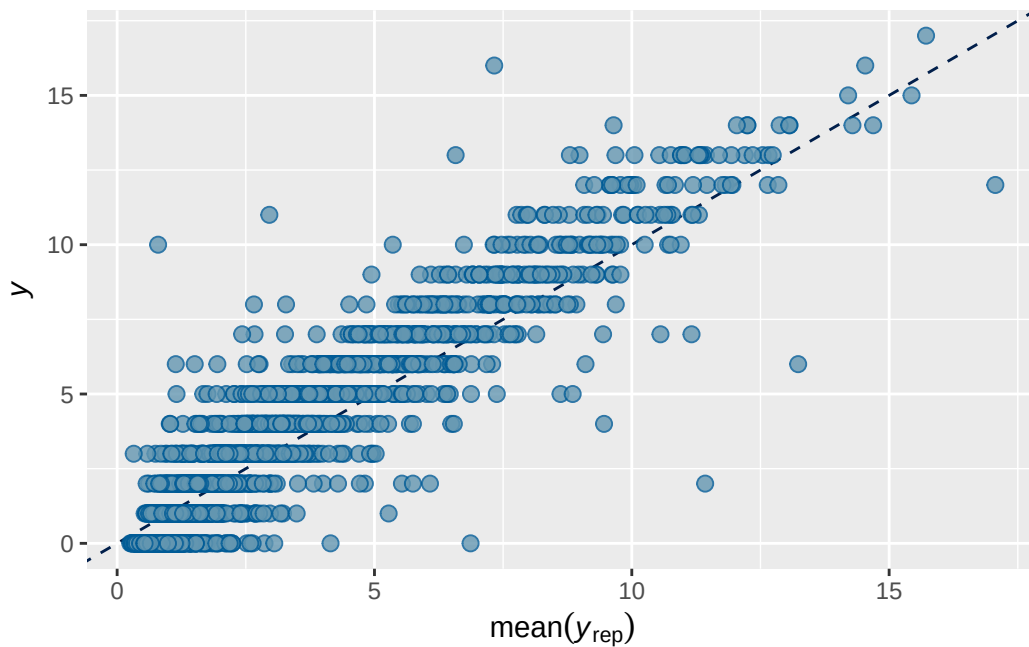


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 1b longitudinal output

```

      Estimate Est.Error Q2.5 Q97.5
R2    0.736    0.00755 0.72  0.75

```

Family: poisson

Links: mu = log

Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in

Data: analytic_df_wide (Number of observations: 4515)

Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;

total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.884	0.021	0.845	0.925	1.002	1206	2087

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.093	0.036	0.023	0.166	1.007	409	563

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	0.191	0.039	0.113	0.266	1.001
breast_development_p_wave_2_sc	0.123	0.039	0.046	0.200	1.000
age_wave_2_sc	-0.105	0.038	-0.179	-0.029	1.000
ethnraceblack_nh	-0.088	0.066	-0.218	0.038	1.000
ethnracehispanic	-0.032	0.058	-0.141	0.078	1.002
ethnraceother	-0.007	0.062	-0.127	0.118	1.000
inr_wave_0_sc	0.087	0.042	0.005	0.169	1.001
adhd_traits_wave_0_sc	1.322	0.033	1.257	1.388	1.002
	Bulk_ESS	Tail_ESS			
Intercept	1303	1645			
breast_development_p_wave_2_sc	1604	2401			
age_wave_2_sc	1616	2631			
ethnraceblack_nh	1171	1736			
ethnracehispanic	1039	1914			
ethnraceother	1104	1705			
inr_wave_0_sc	1359	2268			
adhd_traits_wave_0_sc	1025	1758			

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 1b longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

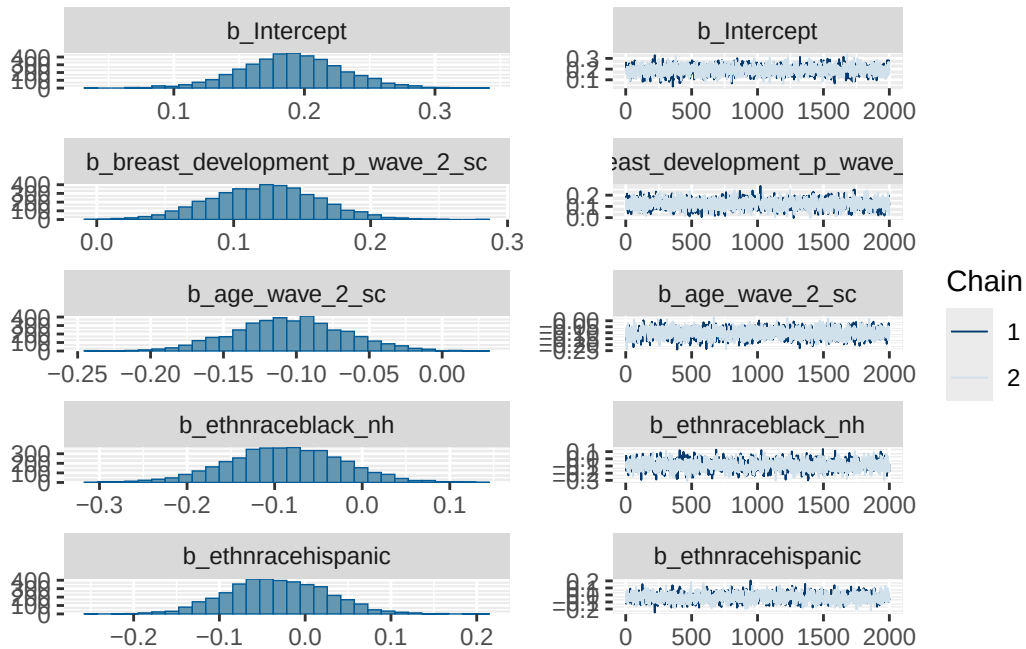
```
# A tibble: 1 x 4
  Variable      Estimate `Lower 95% CI` `Upper 95% CI`
  <chr>      <dbl>      <dbl>      <dbl>
1 Breast development (Wave 2)    9.9         3.62        16.7
```

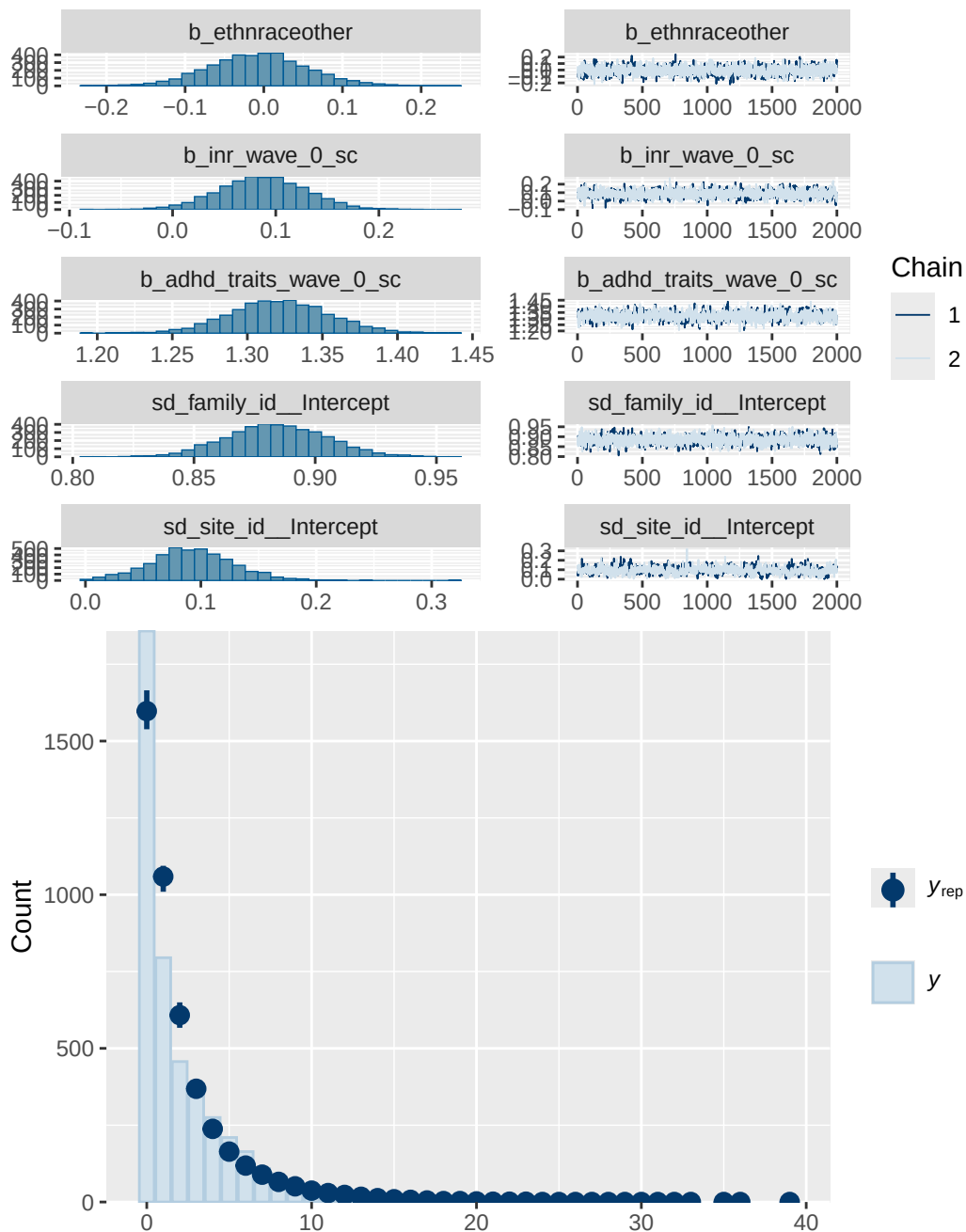
```
[1] 32.7
```

Model 1b longitudinal diagnostics

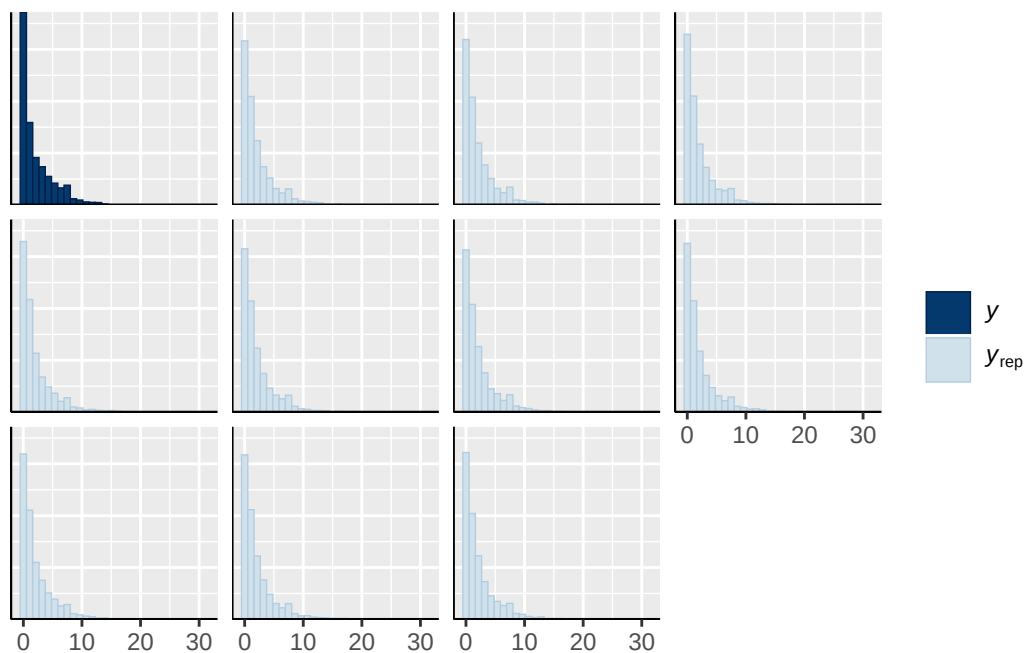
prior	class	coef	group resp
normal(0, 1)	b		
normal(0, 1)	b	adhd_traits_wave_0_sc	
normal(0, 1)	b	age_wave_2_sc	
normal(0, 1)	b	breast_development_p_wave_2_sc	
normal(0, 1)	b	ethnraceblack_nh	

normal(0, 1)	b	ethnracehispanic	
normal(0, 1)	b	ethnraceother	
normal(0, 1)	b	inr_wave_0_sc	
student_t(3, 0, 2.7)	Intercept		family_id
student_t(3, 0, 2.7)	sd	Intercept	family_id
student_t(3, 0, 2.7)	sd		site_id
student_t(3, 0, 2.7)	sd	Intercept	site_id
student_t(3, 0, 2.7)	sd		
dpar nlpar lb ub tag	source		
	user		
	(vectorized)		
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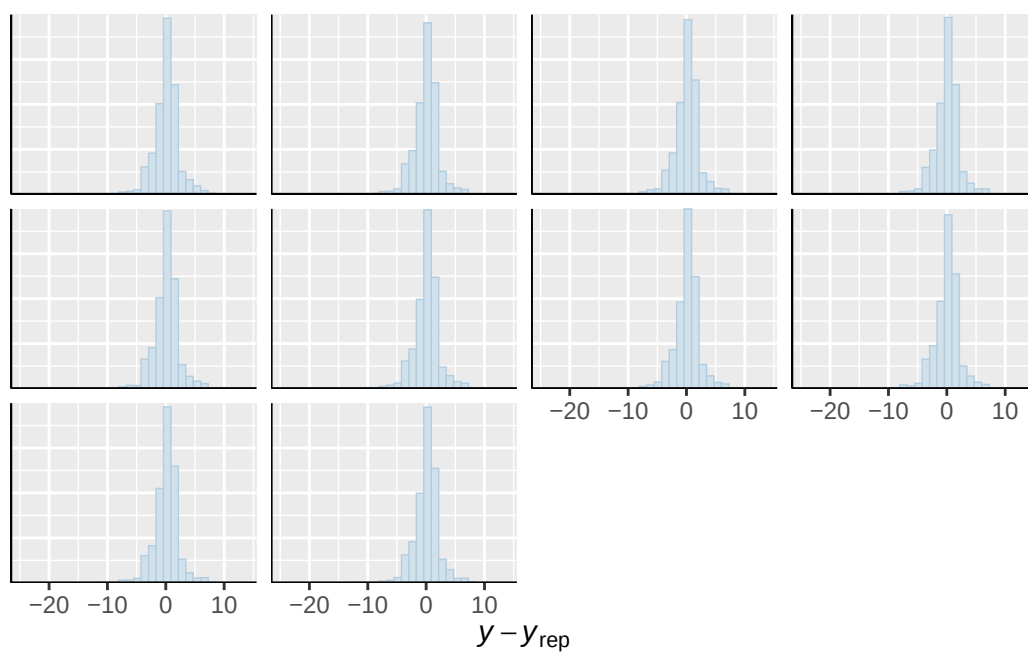




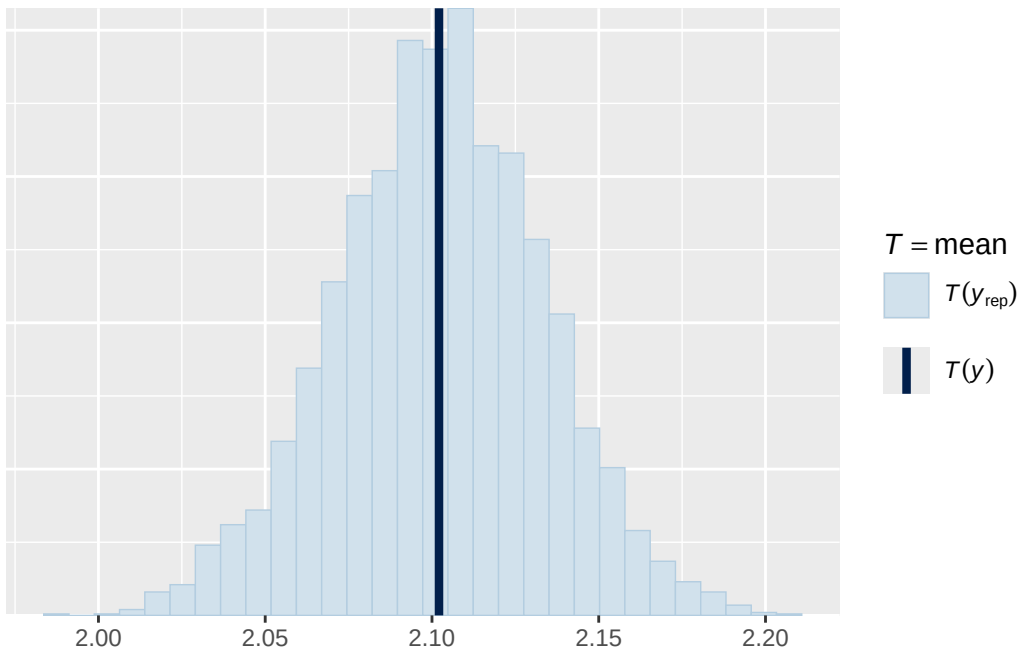
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



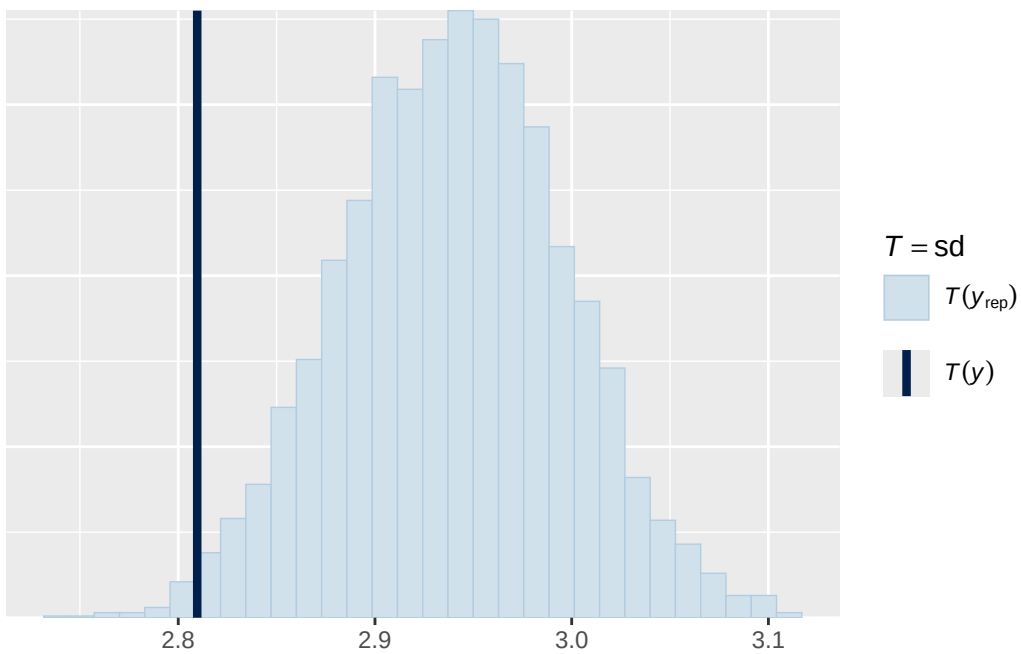
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

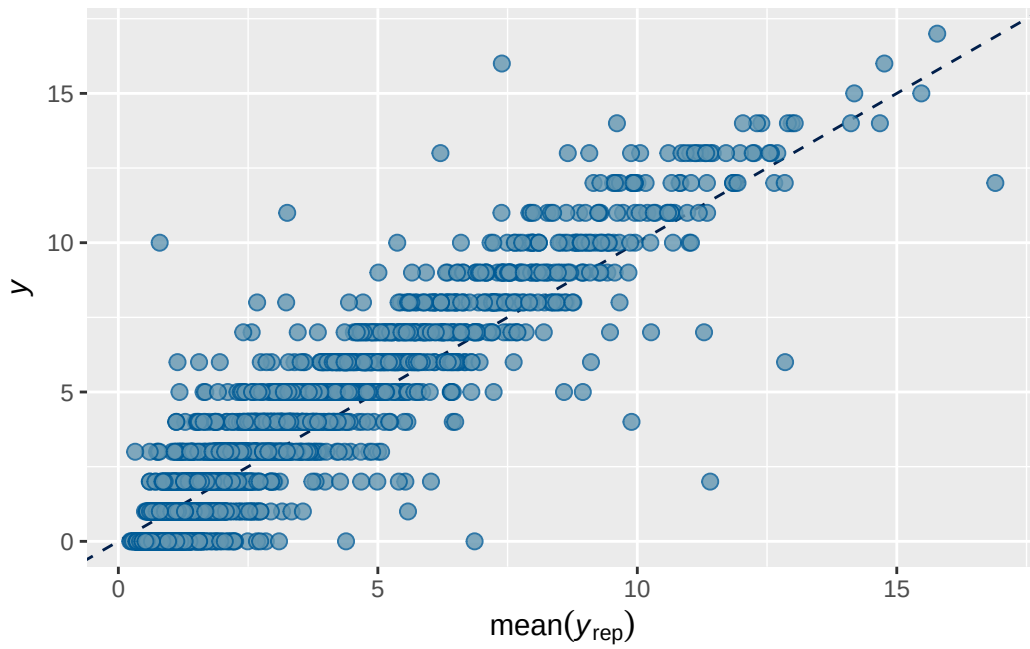


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

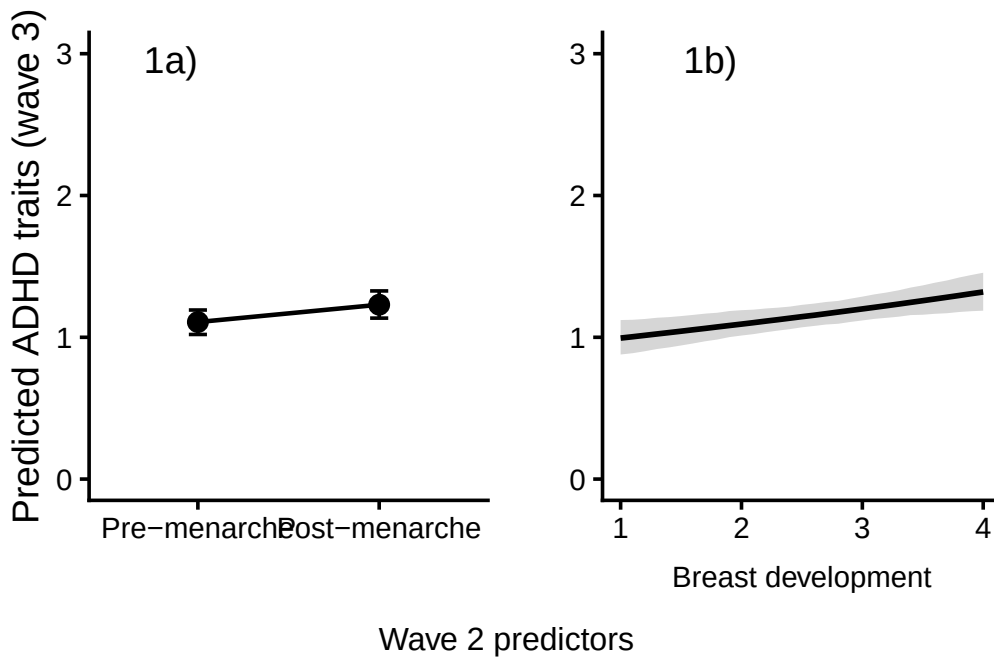


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 1 and 1B Plots



Model 2 longitudinal output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.827      0.00467 0.818 0.836

```

```

Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_life + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3915)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.841    0.015   0.811   0.871 1.002     721    1479

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.110    0.031   0.058   0.177 1.003     600    1161

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI
Intercept          1.188    0.036   1.114
menarche_status_p_wave_2Y    0.183    0.035   0.115
adhd_status_lifeY          0.256    0.042   0.174
age_wave_2_sc        -0.017    0.031  -0.079
ethnraceblack_nh       -0.406    0.056  -0.513
ethnracehispanic       -0.092    0.047  -0.185
ethnraceother          0.003    0.053  -0.101
inr_wave_0_sc          0.082    0.034   0.015
internalising_wave_0_sc    0.835    0.028   0.781
menarche_status_p_wave_2Y:adhd_status_lifeY -0.036    0.064  -0.167
      u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept      1.258 1.002     983    2026
menarche_status_p_wave_2Y    0.249 1.004    1062    1625
adhd_status_lifeY    0.337 1.003    1104    1899
age_wave_2_sc        0.042 1.002    1078    1834
ethnraceblack_nh     -0.297 1.002     883    1596
ethnracehispanic      0.003 1.004     913    1608
ethnraceother         0.105 1.004     665    1648
inr_wave_0_sc         0.150 1.002    1158    1768

```

internalising_wave_0_sc	0.889	1.000	789	1376
menarche_status_p_wave_2Y:adhd_status_lifeY	0.085	1.003	876	1750

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2 longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

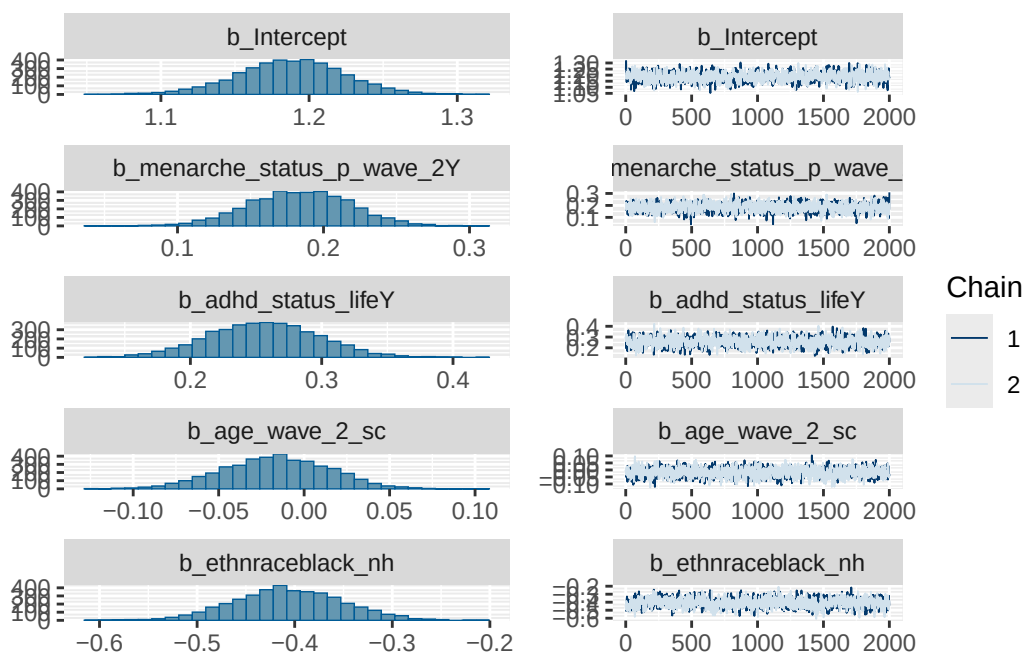
A tibble: 3 x 4

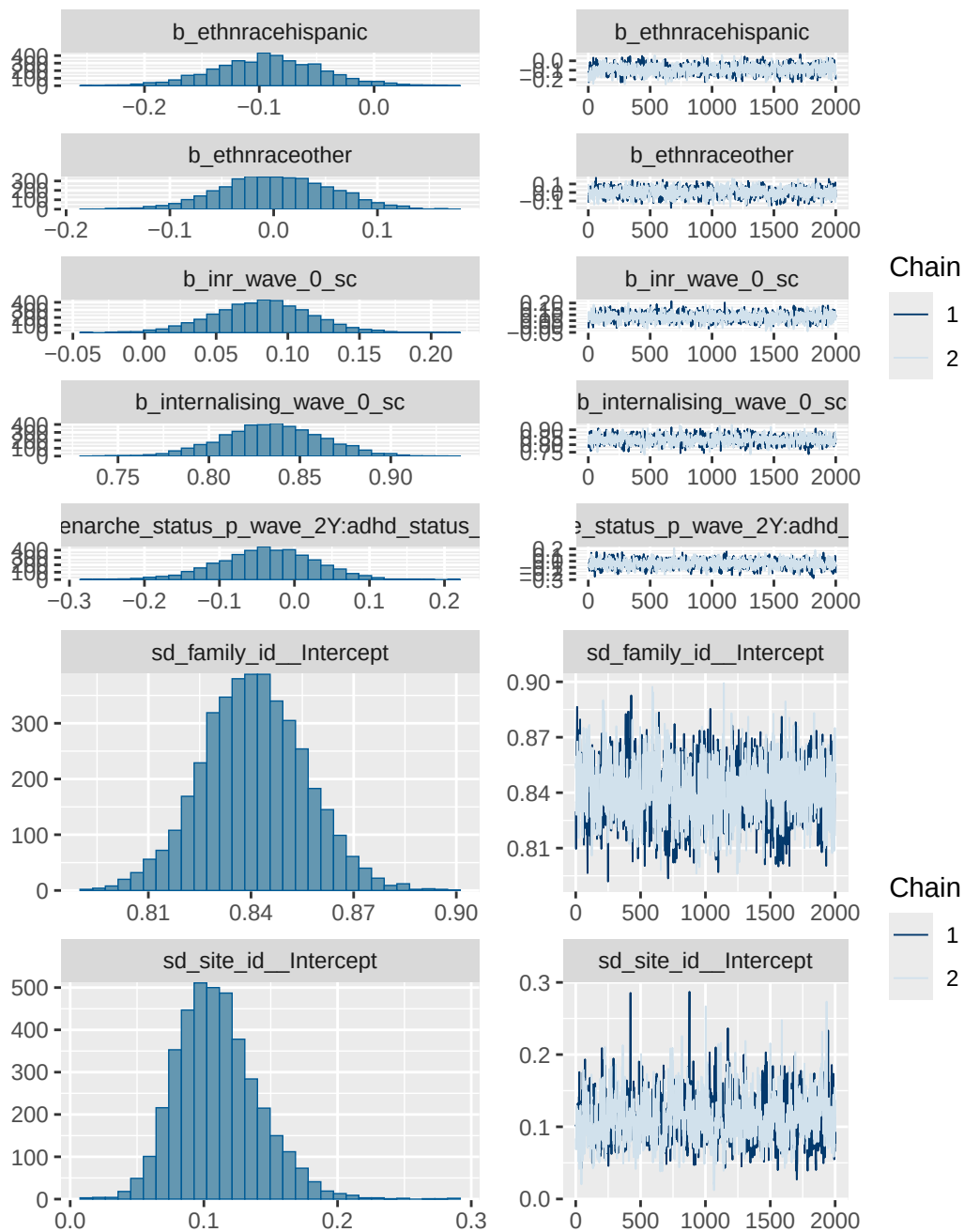
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche Y (Wave 2) [ADHD = 0]	20.1	12.2	28.3
2 ADHD Y [Menarche = 0]	29.2	19.0	40.1
3 Menarche × ADHD	-3.58	-15.4	8.83

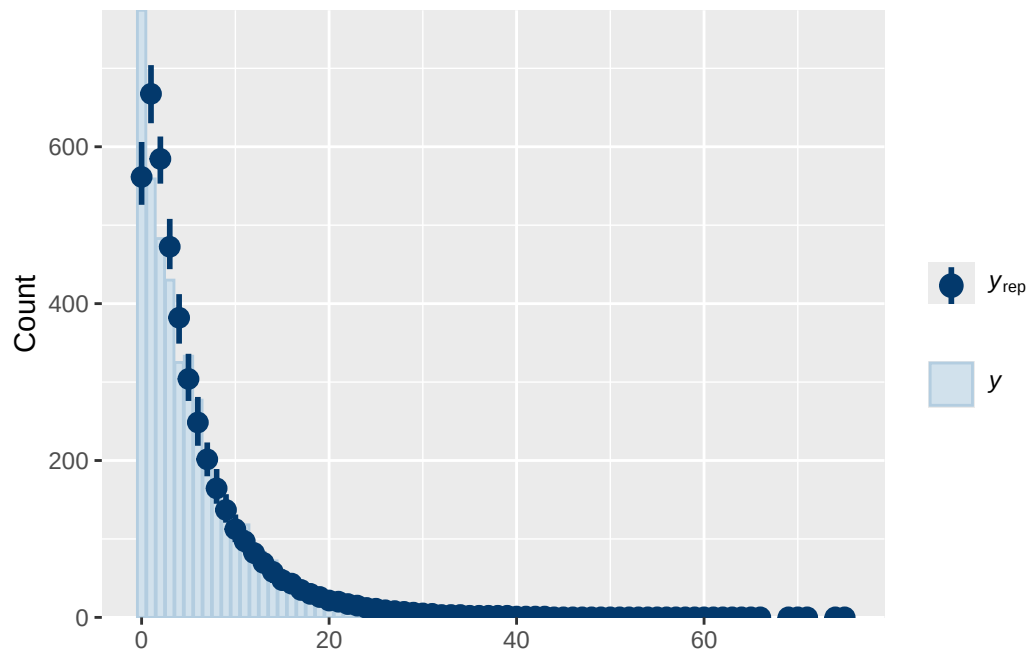
Model 2 longitudinal diagnostics

prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_status_lifeY
normal(0, 1)	b	age_wave_2_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	inr_wave_0_sc
normal(0, 1)	b	internalising_wave_0_sc
normal(0, 1)	b	menarche_status_p_wave_2Y
normal(0, 1)	b	menarche_status_p_wave_2Y:adhd_status_lifeY
student_t(3, 1.4, 2.5)	Intercept	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag	source	
	user	
	(vectorized)	
	(vectorized)	
	(vectorized)	

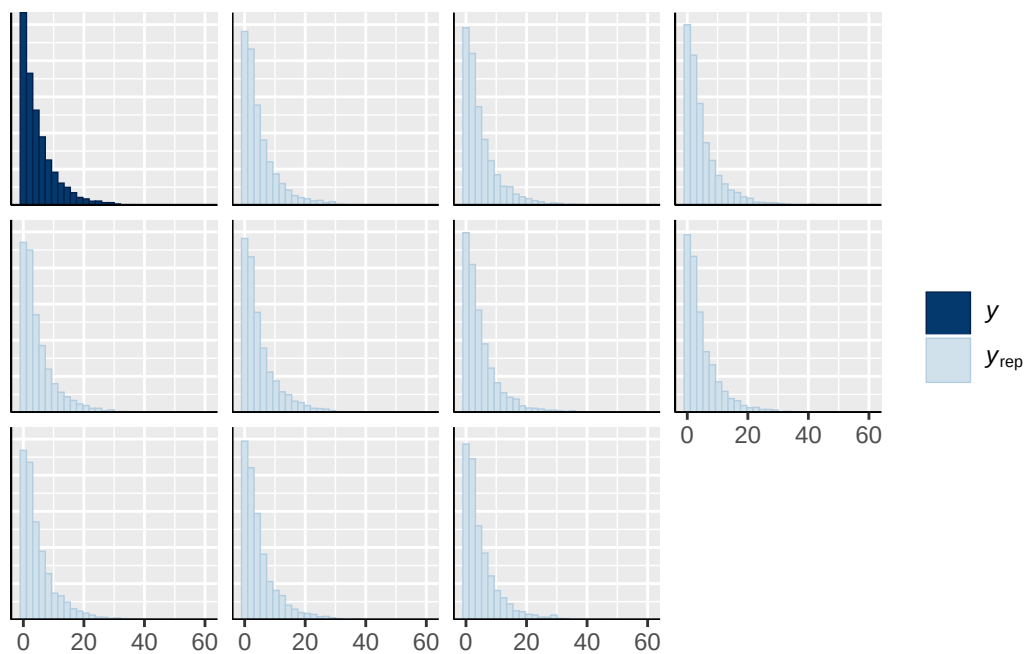
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		(vectorized)
		(vectorized)
		(vectorized)
	default	
	default	
family_id	0	(vectorized)
family_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



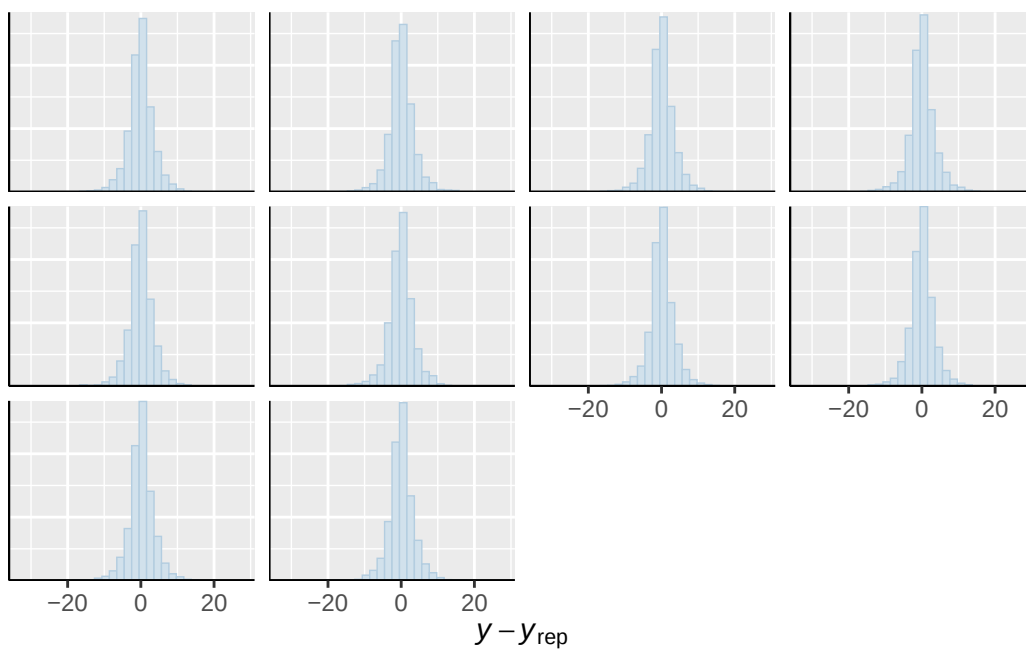




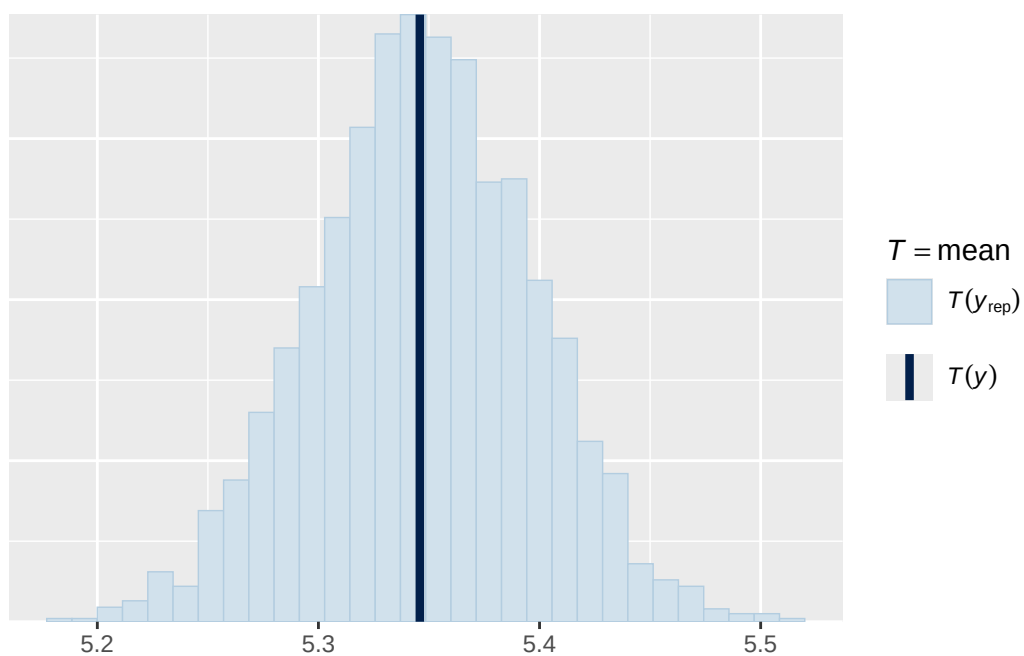
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



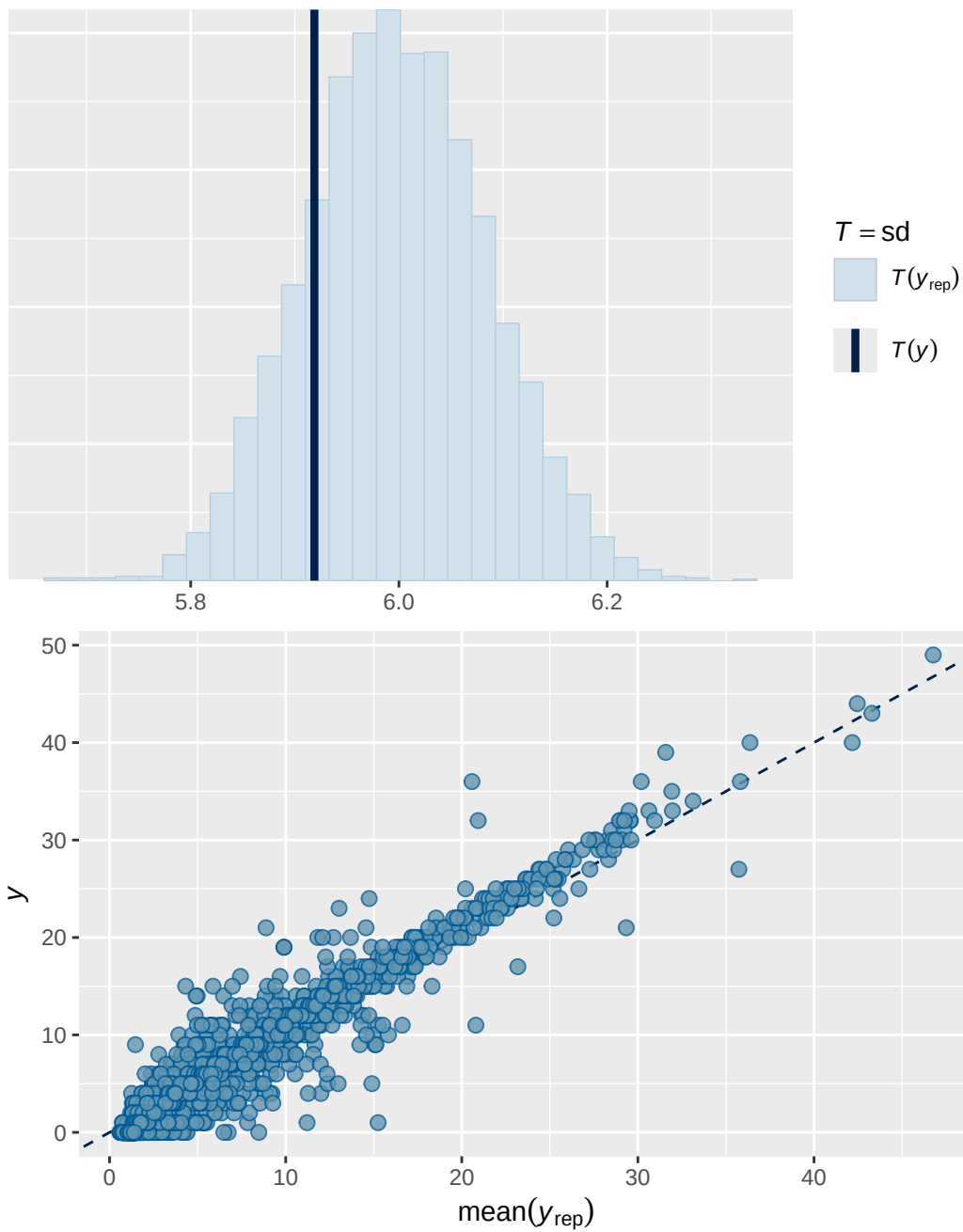
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 2b longitudinal output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.827	0.00448	0.818	0.836

Family: poisson
 Links: mu = log
 Formula: internalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_status_life + age_wave_2_sc
 Data: analytic_df_wide (Number of observations: 4515)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.845	0.015	0.815	0.875	1.001	759	1765

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.113	0.031	0.062	0.184	1.000	733	1372

Regression Coefficients:

	Estimate	Est.Error	1-95% CI
Intercept	1.266	0.035	1.196
breast_development_p_wave_2_sc	0.150	0.032	0.088
adhd_status_lifeY	0.225	0.034	0.160
age_wave_2_sc	0.013	0.029	-0.042
ethnraceblack_nh	-0.390	0.054	-0.499
ethnracehispanic	-0.076	0.048	-0.169
ethnraceother	-0.009	0.051	-0.108
inr_wave_0_sc	0.063	0.034	-0.002
internalising_wave_0_sc	0.839	0.028	0.786
breast_development_p_wave_2_sc:adhd_status_lifeY	0.016	0.067	-0.121

	u-95% CI	Rhat	Bulk_ESS
Intercept	1.334	1.003	1109
breast_development_p_wave_2_sc	0.212	1.002	1465
adhd_status_lifeY	0.291	1.001	1106
age_wave_2_sc	0.071	1.002	1408
ethnraceblack_nh	-0.284	1.001	1157
ethnracehispanic	0.017	1.002	793
ethnraceother	0.090	1.001	758
inr_wave_0_sc	0.131	1.002	1131
internalising_wave_0_sc	0.893	1.004	671
breast_development_p_wave_2_sc:adhd_status_lifeY	0.142	1.000	1219

	Tail_ESS
Intercept	1831
breast_development_p_wave_2_sc	1863
adhd_status_lifeY	1852

age_wave_2_sc	2115
ethnraceblack_nh	2080
ethnracehispanic	1431
ethnraceother	1832
inr_wave_0_sc	1760
internalising_wave_0_sc	784
breast_development_p_wave_2_sc:adhd_status_lifeY	1772

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2b longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

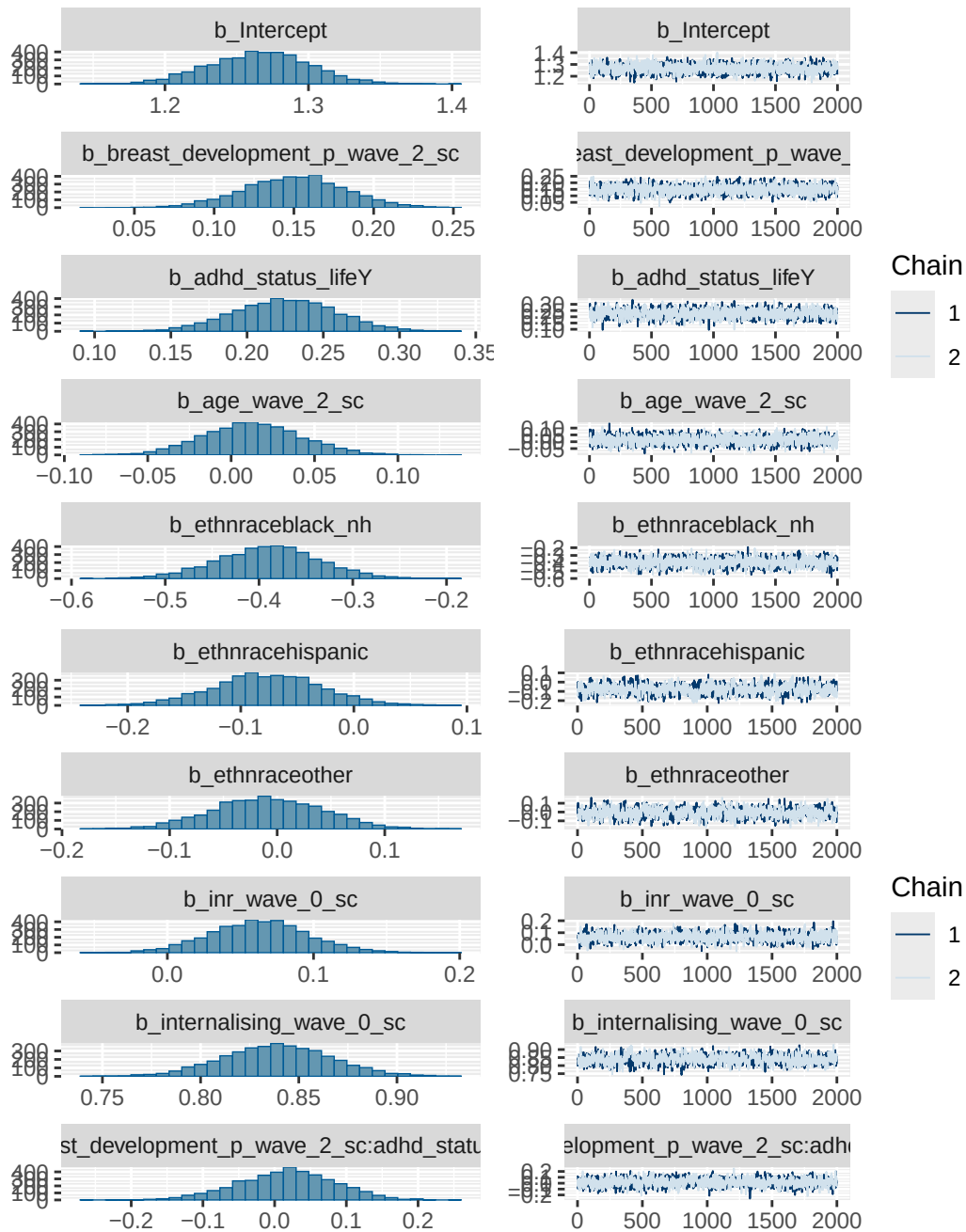
```
# A tibble: 3 x 4
```

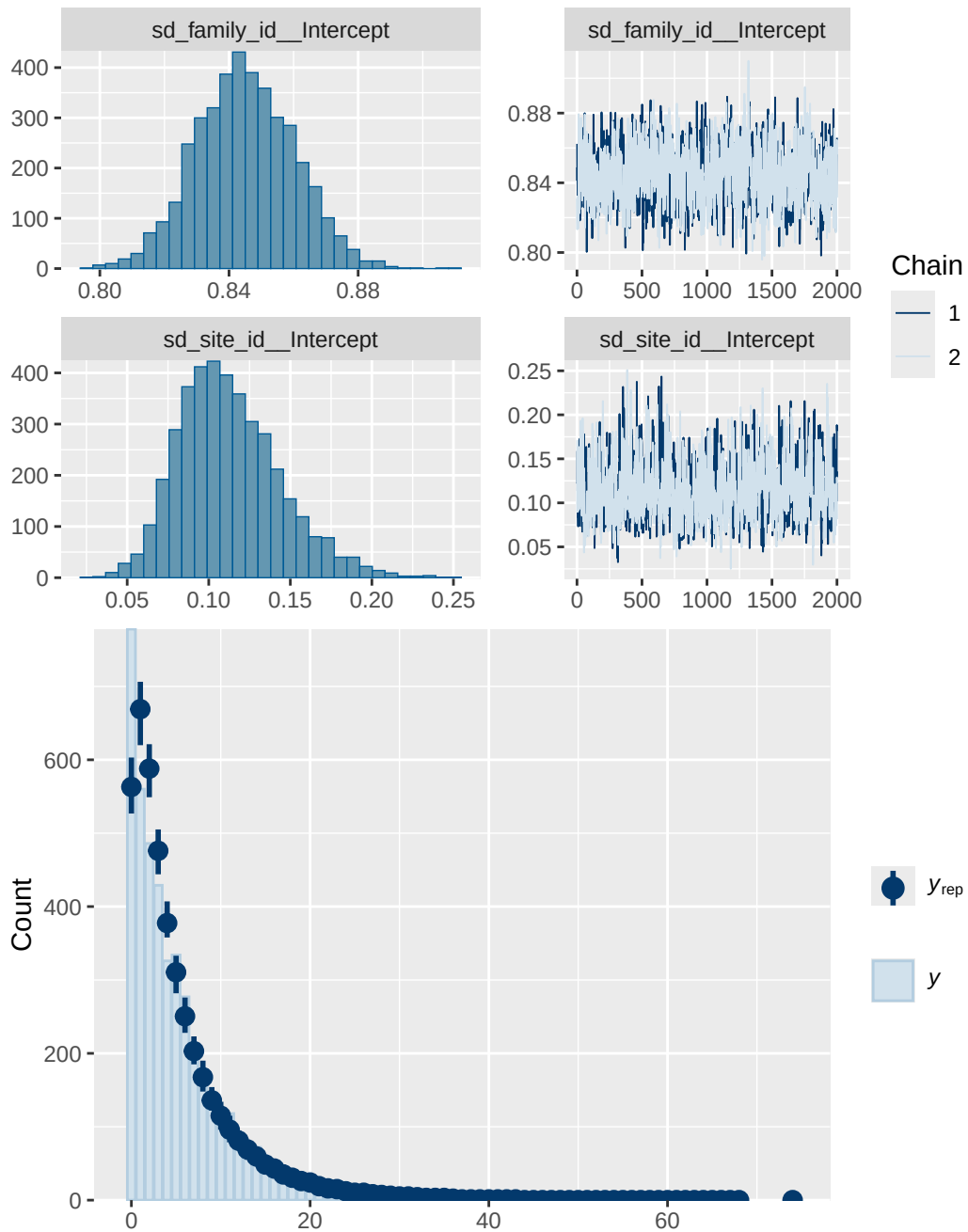
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (Wave 2) [ADHD = 0]	12.3	7.03	17.8
2 ADHD Y [Breast development = 0]	25.2	17.3	33.7
3 Breast development × ADHD	1.27	-8.9	11.5

Model 2b longitudinal diagnostics

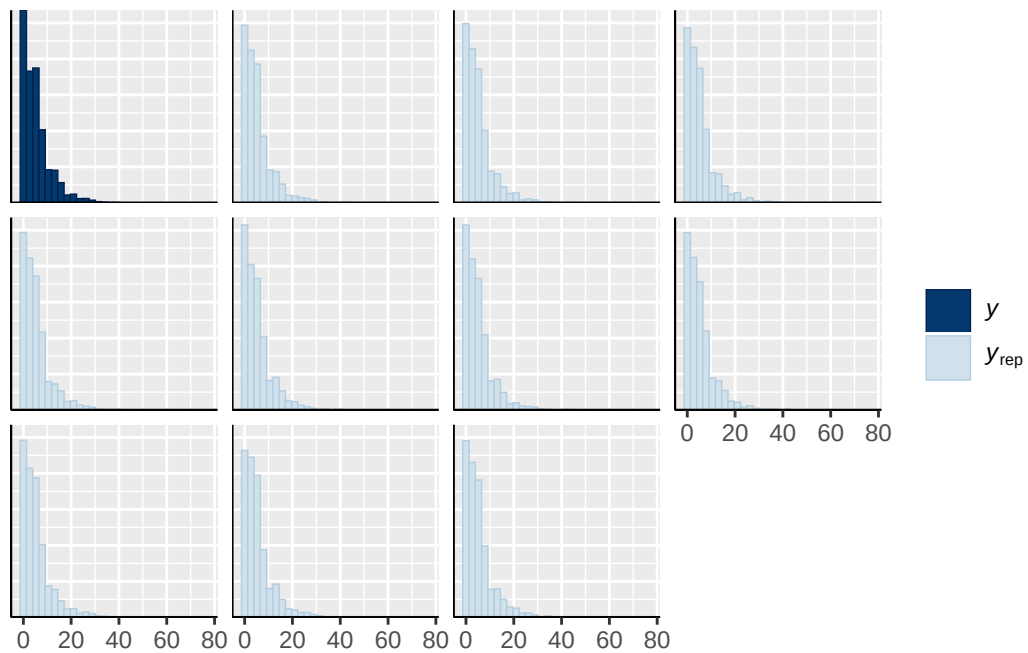
prior	class
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
student_t(3, 1.4, 2.5)	Intercept
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd

		coef	group	resp	dpar	nlpar	lb
		adhd_status_lifeY					
		age_wave_2_sc					
		breast_development_p_wave_2_sc					
		breast_development_p_wave_2_sc:adhd_status_lifeY					
		ethnraceblack_nh					
		ethnracehispanic					
		ethnraceother					
		inr_wave_0_sc					
		internalising_wave_0_sc					
							0
			family_id				0
		Intercept	family_id				0
			site_id				0
		Intercept	site_id				0
ub tag	source						
	user						
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	default						
	default						
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	(vectorized)						
	(vectorized)						

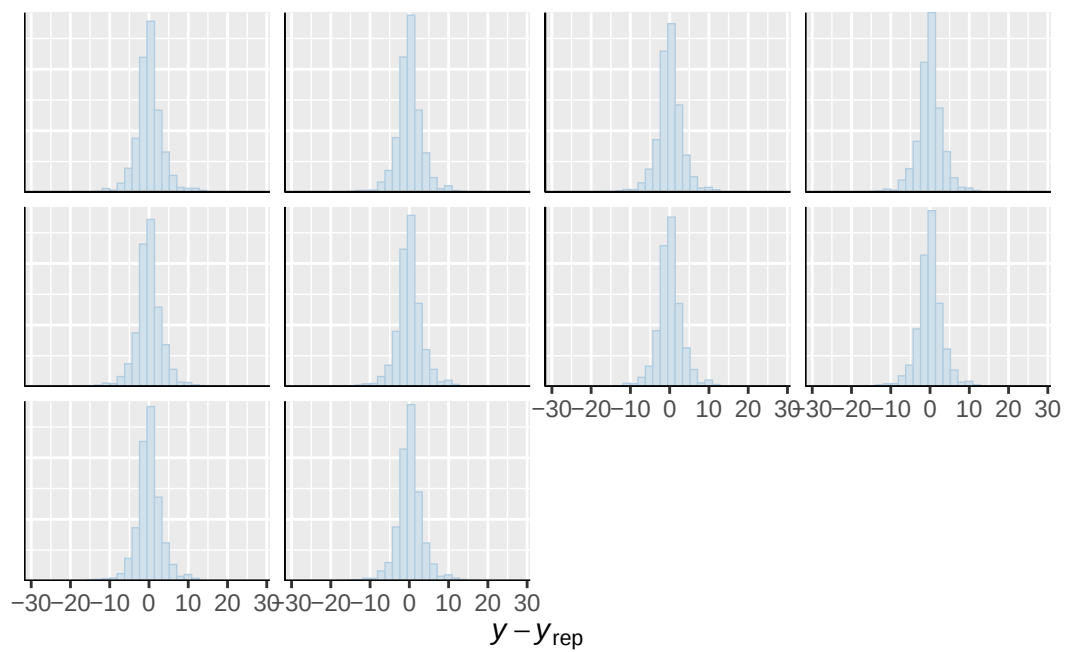




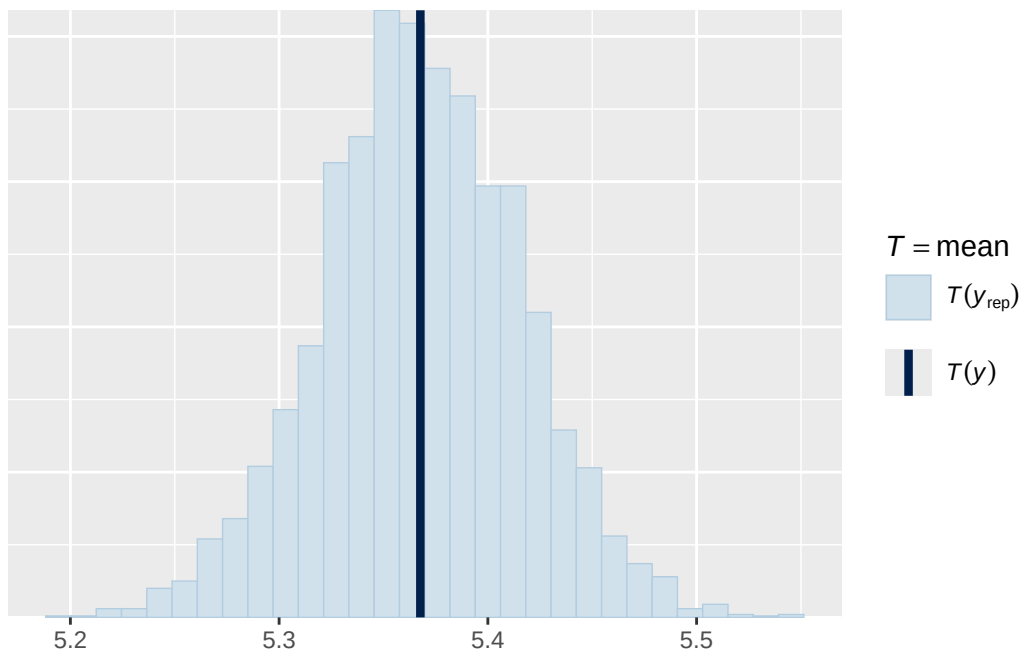
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



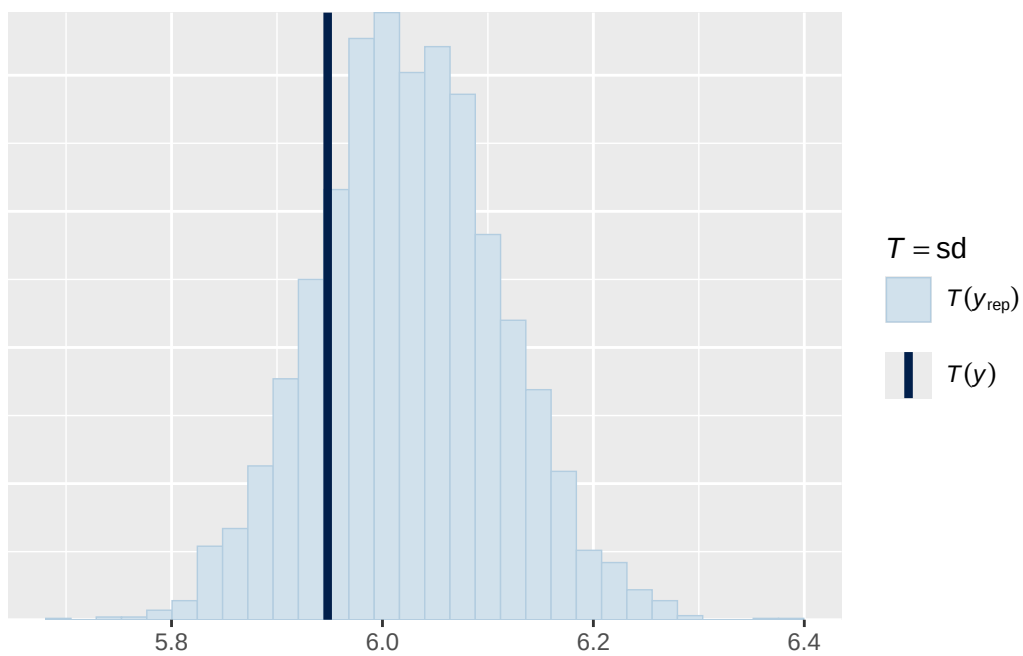
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

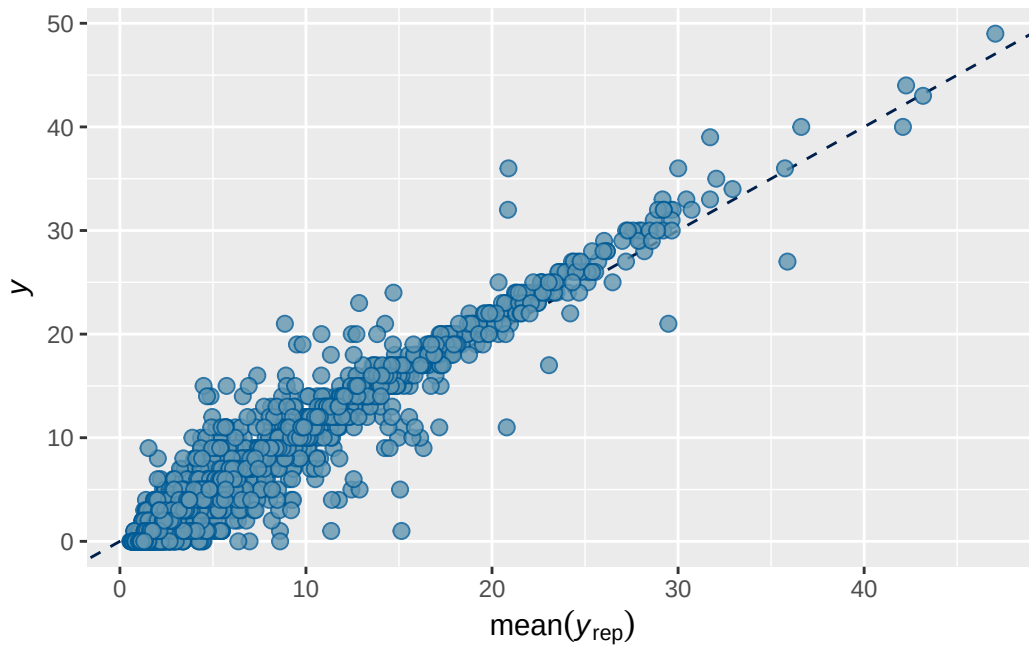


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 3 longitudinal output

```

      Estimate Est.Error  Q2.5 Q97.5
R2    0.823      0.0053 0.813 0.833

```

Family: poisson

Links: mu = log

Formula: externalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_life + age_wave_2_sc

Data: analytic_df_wide (Number of observations: 4496)

Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;

total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.013	0.020	0.974	1.053	1.001	2352	4260

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.061	0.037	0.004	0.140	1.003	638	2023

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	
Intercept	0.349	0.040	0.273	
menarche_status_p_wave_2Y	0.109	0.046	0.018	
adhd_status_lifeY	0.547	0.053	0.443	
age_wave_2_sc	-0.101	0.040	-0.178	
ethnraceblack_nh	-0.097	0.066	-0.228	
ethnracehispanic	-0.088	0.058	-0.201	
ethnraceother	-0.033	0.064	-0.158	
inr_wave_0_sc	0.002	0.044	-0.083	
externalising_wave_0_sc	0.964	0.034	0.899	
menarche_status_p_wave_2Y:adhd_status_lifeY	0.088	0.079	-0.070	
	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.425	1.000	4405	5803
menarche_status_p_wave_2Y	0.199	1.001	4433	6005
adhd_status_lifeY	0.651	1.001	3805	5303
age_wave_2_sc	-0.020	1.000	4048	5123
ethnraceblack_nh	0.031	1.001	2902	4425
ethnracehispanic	0.027	1.000	3041	4770
ethnraceother	0.092	1.000	3116	3970
inr_wave_0_sc	0.090	1.000	3209	4833
externalising_wave_0_sc	1.031	1.000	2641	4461
menarche_status_p_wave_2Y:adhd_status_lifeY	0.241	1.002	3454	4871

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 3 longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

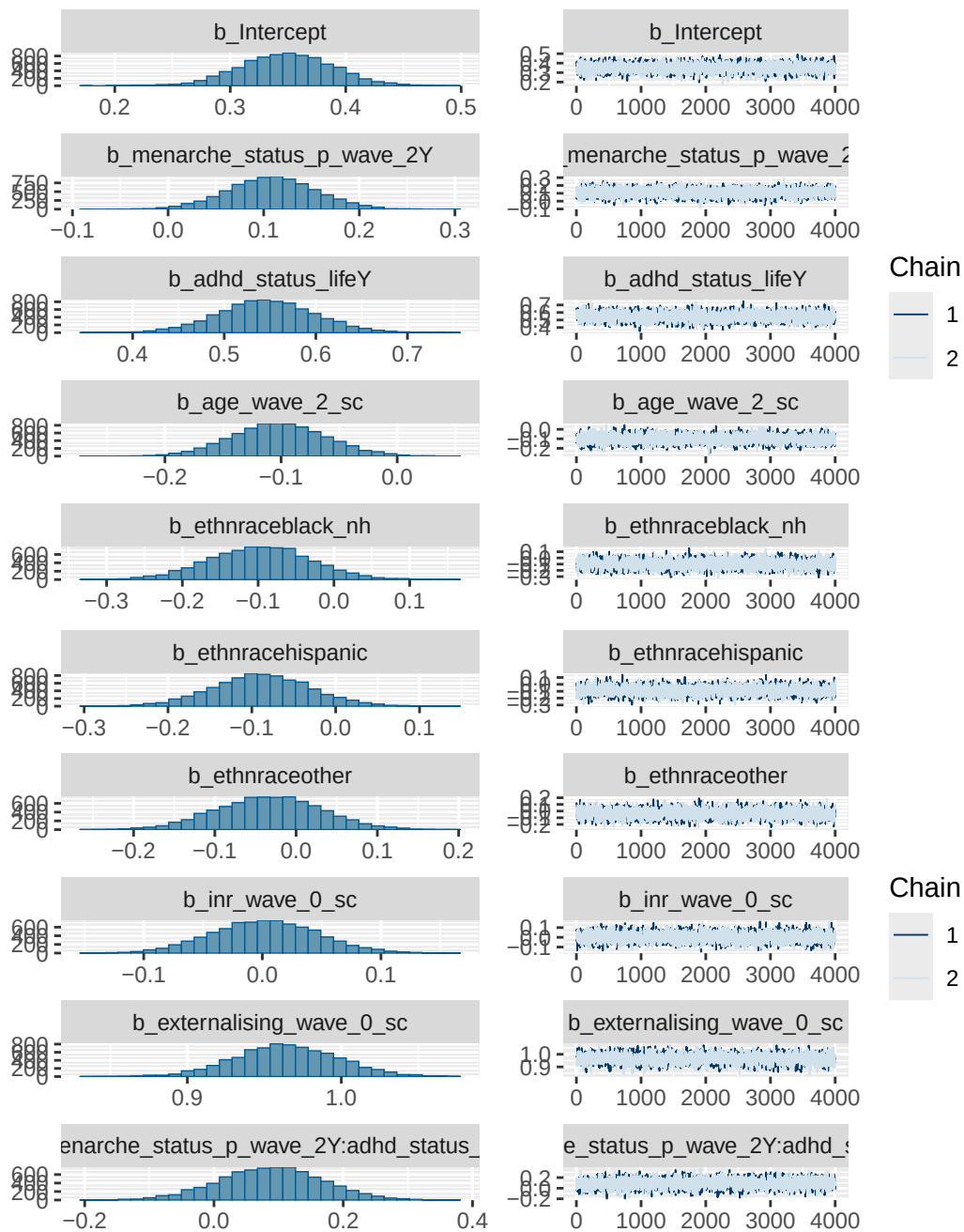
```
# A tibble: 3 x 4
```

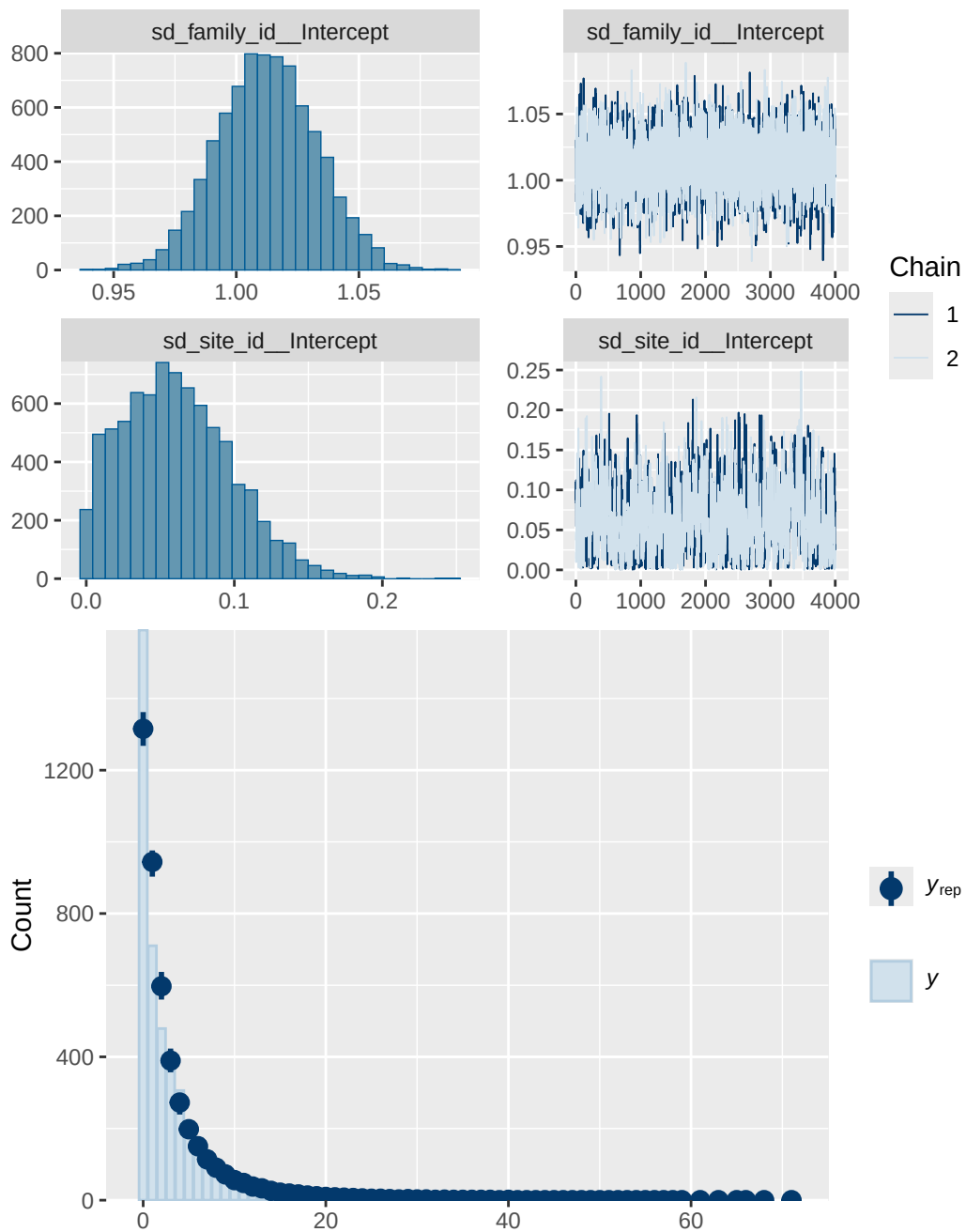
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche Y (Wave 2) [ADHD = 0]	11.5	1.81	22.0
2 ADHD Y [Menarche = 0]	72.7	55.7	91.8
3 Menarche × ADHD	9.2	-6.77	27.3

Model 3 longitudinal diagnostics

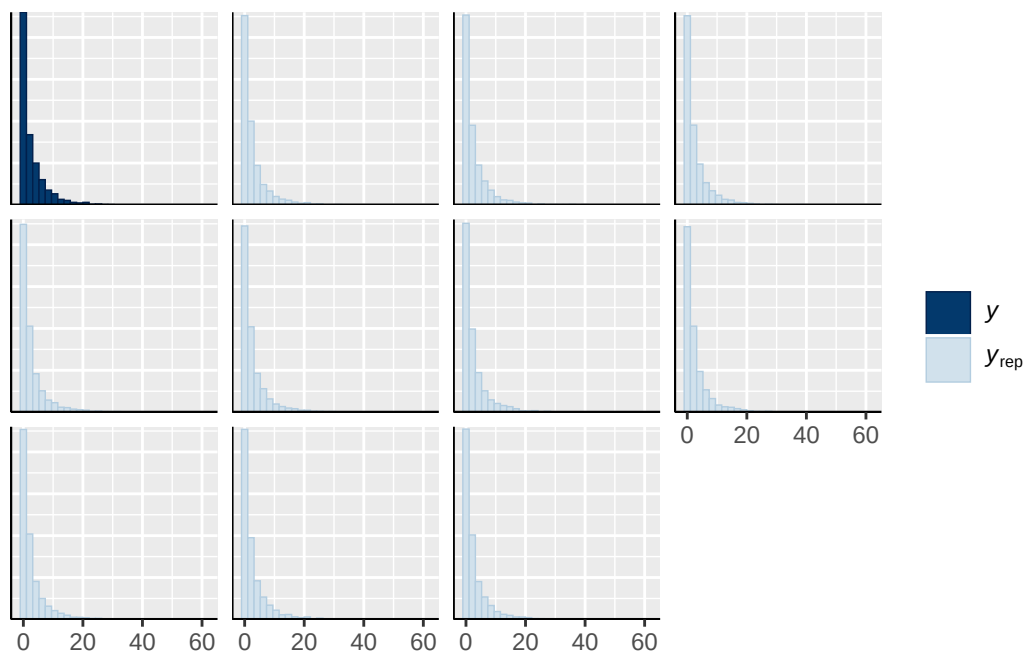
prior	class	coef
normal(0, 1)	b	

normal(0, 1)	b	adhd_status_lifeY
normal(0, 1)	b	age_wave_2_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	externalising_wave_0_sc
normal(0, 1)	b	inr_wave_0_sc
normal(0, 1)	b	menarche_status_p_wave_2Y
normal(0, 1)	b	menarche_status_p_wave_2Y:adhd_status_lifeY
student_t(3, 0, 2.9)	Intercept	
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
group resp dpar nlpar lb ub tag	source	
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	default	
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

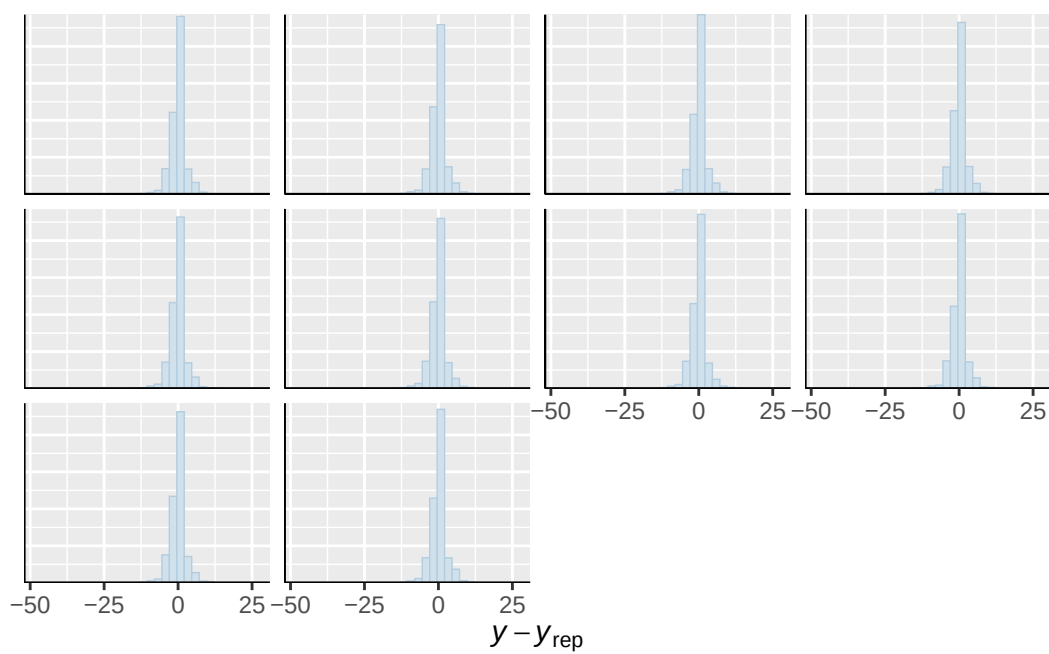




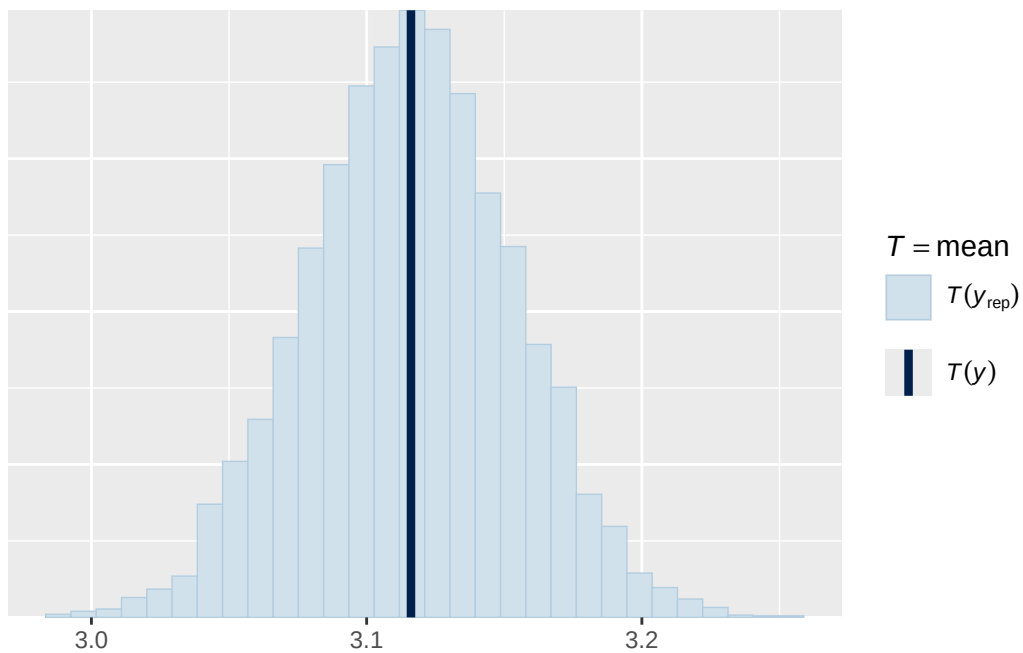
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



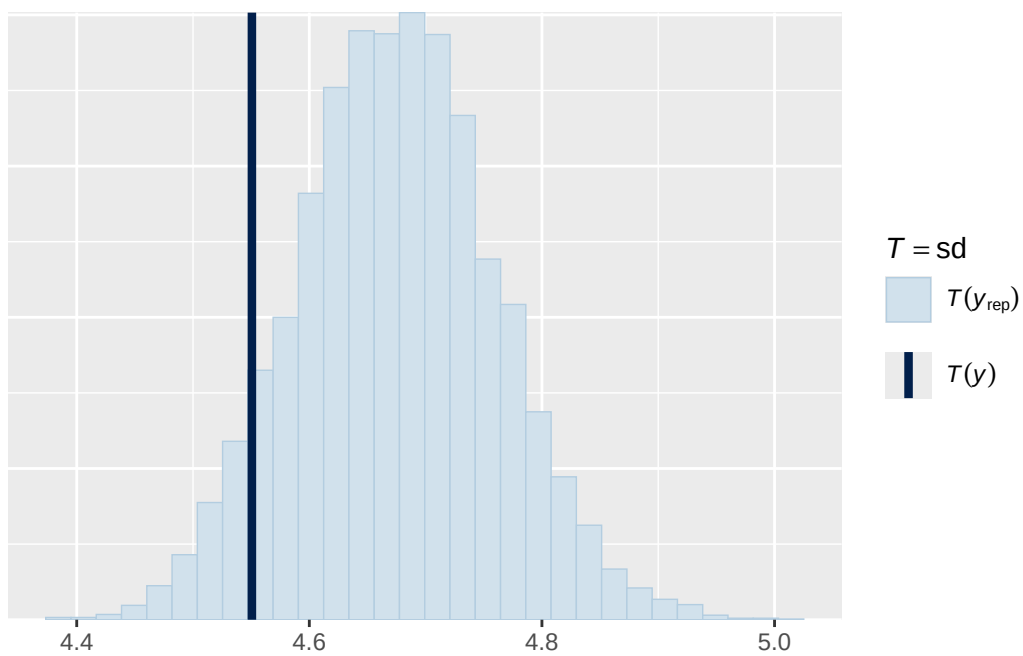
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

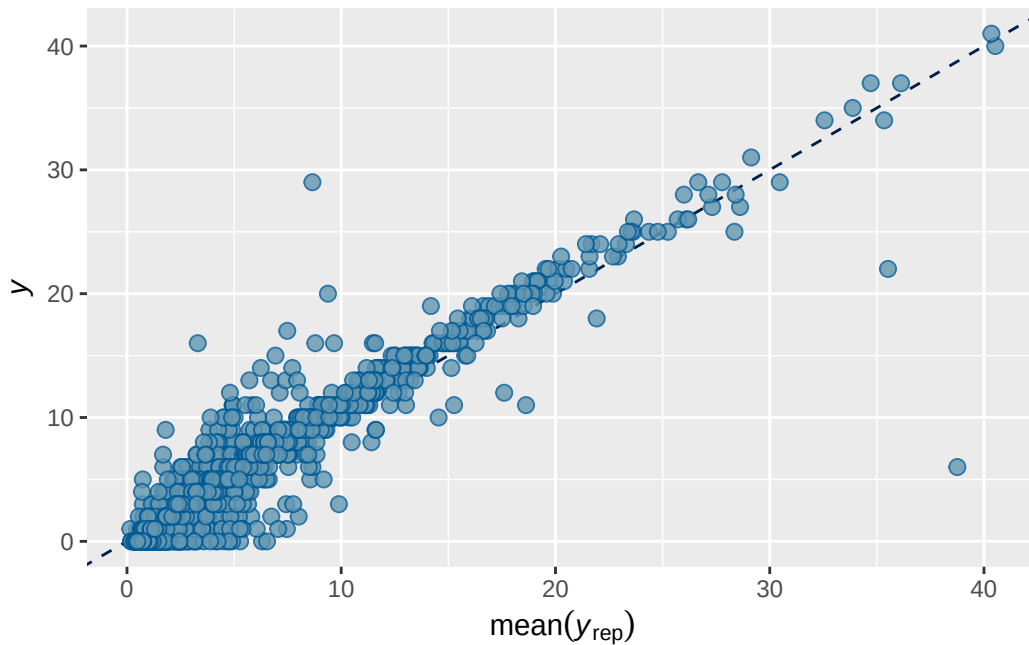


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 3b longitudinal output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.825      0.00514 0.815 0.835

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_status_life + age_wave
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.018	0.020	0.978	1.059	1.000	1968	4079

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.060	0.035	0.004	0.136	1.002	659	2137

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	
Intercept	0.398	0.036	0.328	
breast_development_p_wave_2_sc	0.171	0.044	0.084	
adhd_status_lifeY	0.562	0.043	0.477	
age_wave_2_sc	-0.106	0.038	-0.183	
ethnraceblack_nh	-0.108	0.068	-0.240	
ethnracehispanic	-0.080	0.057	-0.195	
ethnraceother	-0.051	0.066	-0.180	
inr_wave_0_sc	-0.013	0.044	-0.099	
externalising_wave_0_sc	0.964	0.034	0.897	
breast_development_p_wave_2_sc:adhd_status_lifeY	0.047	0.084	-0.122	
	u-95% CI	Rhat	Bulk_ESS	
Intercept	0.467	1.000	3922	
breast_development_p_wave_2_sc	0.256	1.001	4900	
adhd_status_lifeY	0.647	1.001	3524	
age_wave_2_sc	-0.033	1.003	3708	
ethnraceblack_nh	0.026	1.000	2958	
ethnracehispanic	0.032	1.000	3219	
ethnraceother	0.078	1.001	3096	
inr_wave_0_sc	0.077	1.001	2985	
externalising_wave_0_sc	1.031	1.000	2442	
breast_development_p_wave_2_sc:adhd_status_lifeY	0.211	1.000	3712	
	Tail_ESS			
Intercept	4915			
breast_development_p_wave_2_sc	6012			
adhd_status_lifeY	4915			
age_wave_2_sc	4880			
ethnraceblack_nh	4267			
ethnracehispanic	4612			
ethnraceother	4817			
inr_wave_0_sc	4795			
externalising_wave_0_sc	3957			
breast_development_p_wave_2_sc:adhd_status_lifeY	5296			

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

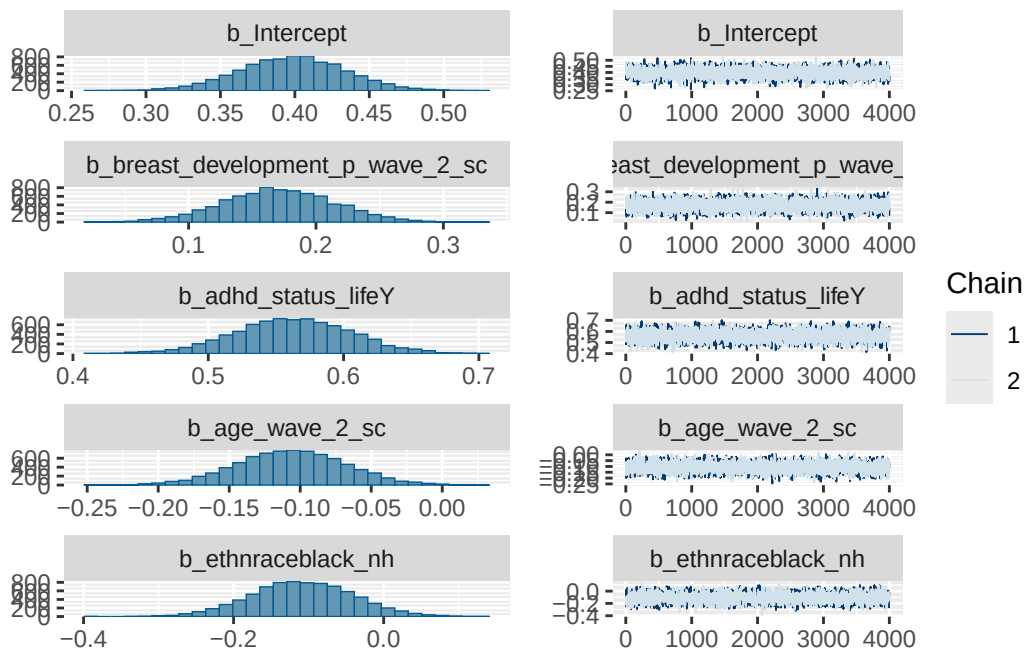
Model 3b longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

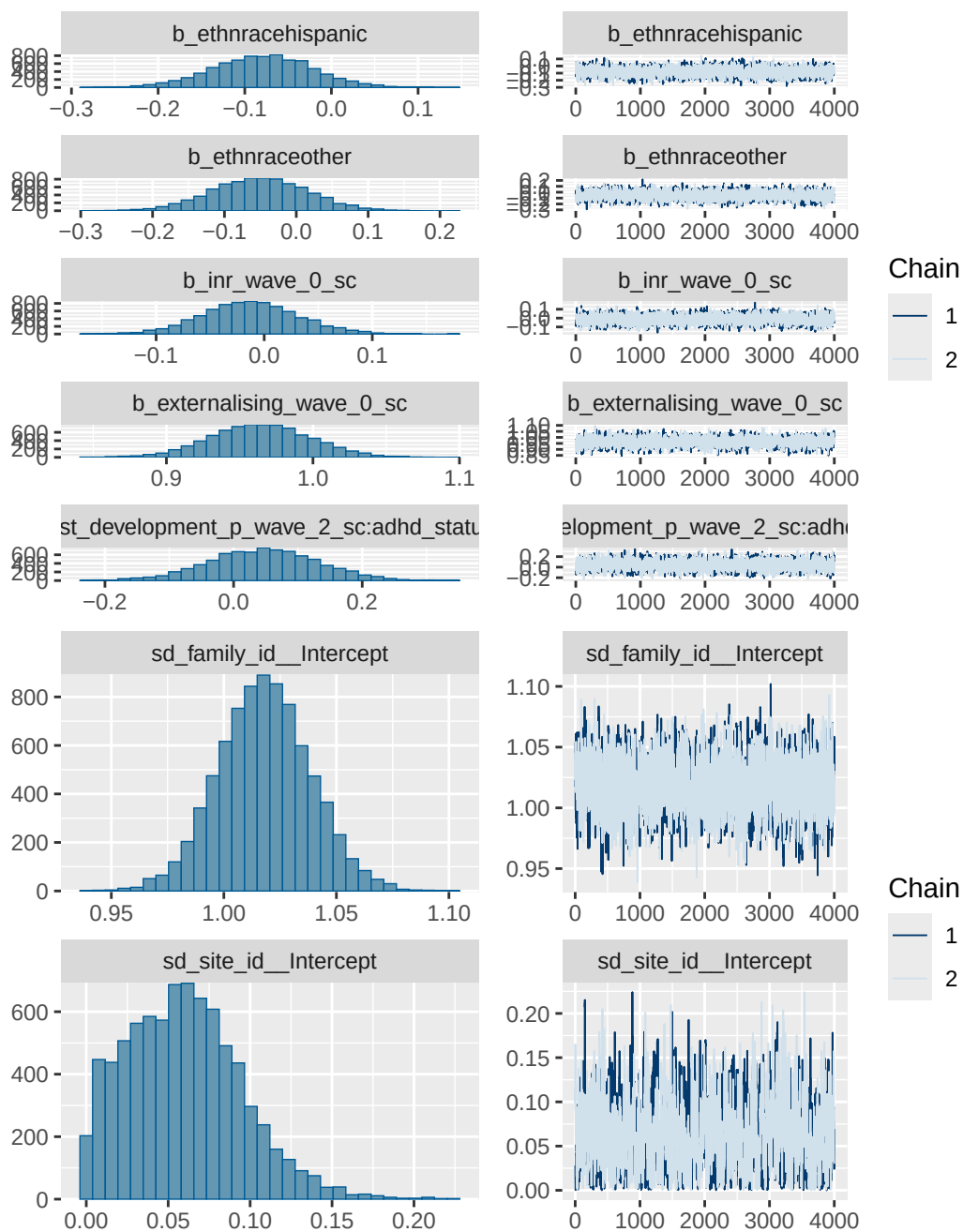
A tibble: 3 x 4

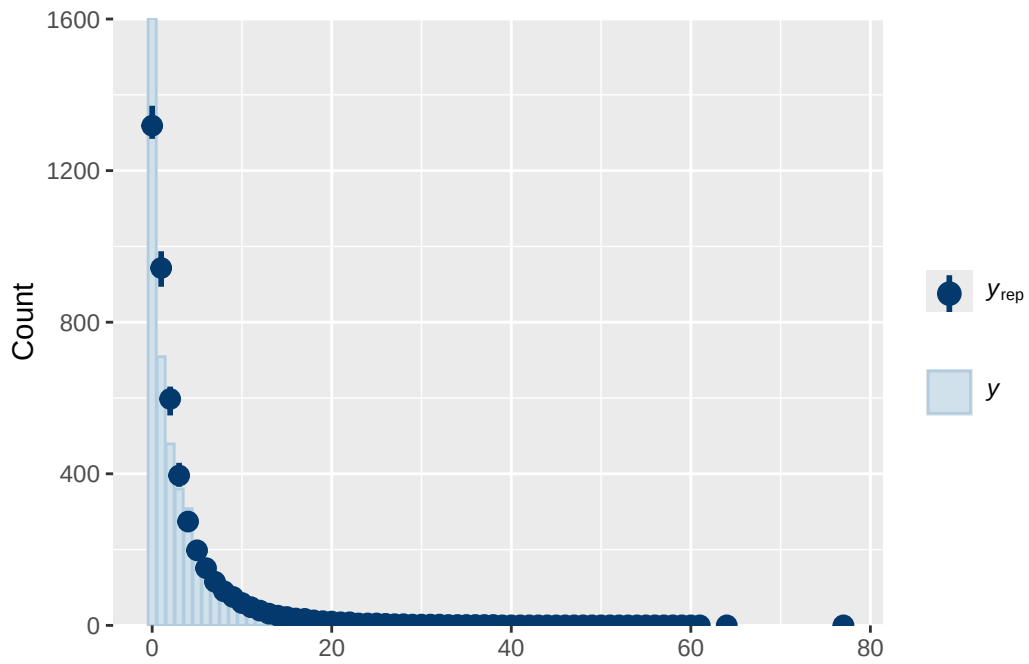

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ub tag      source
          user
          (vectorized)
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          default
          default
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          (vectorized)

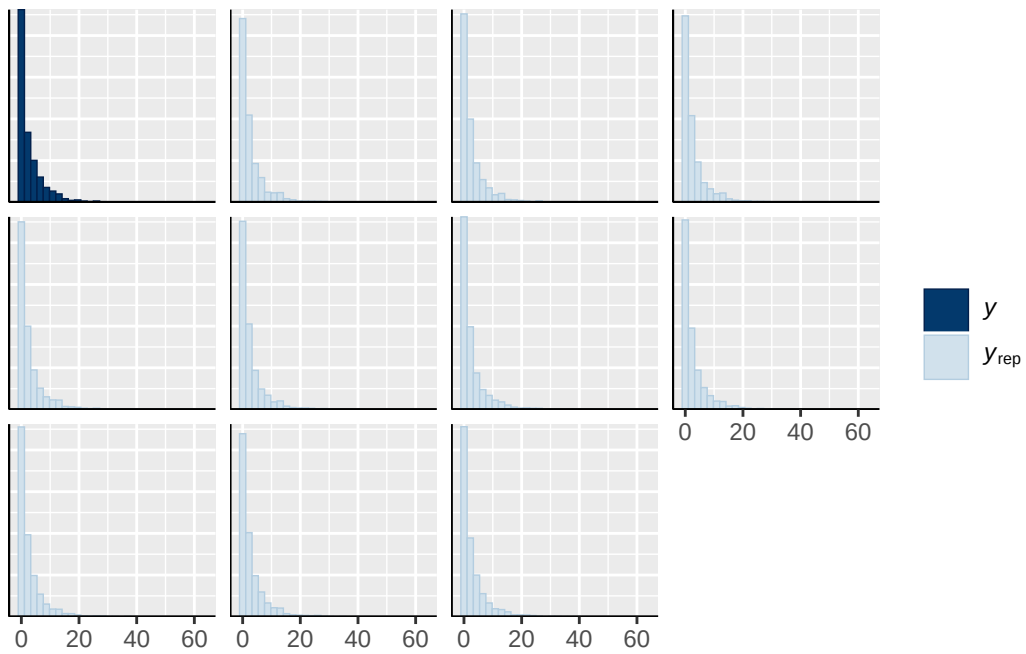
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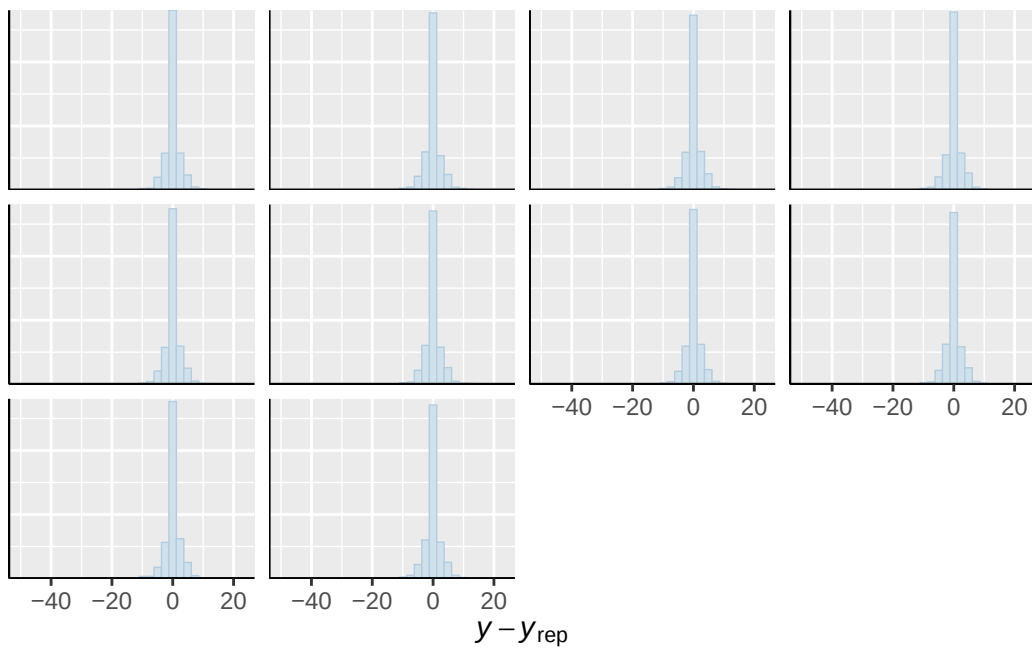




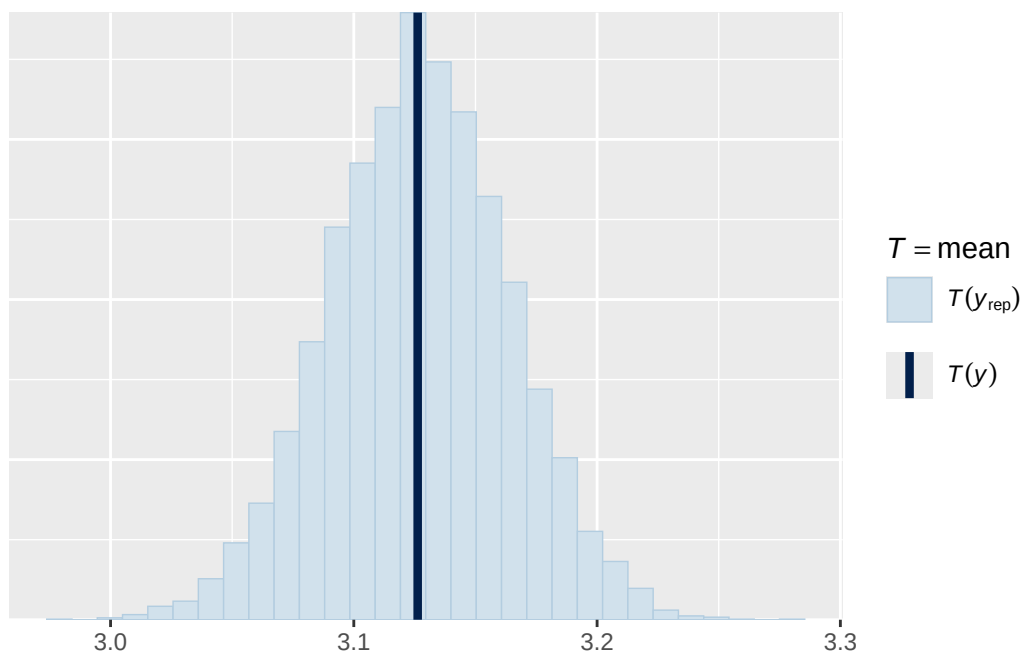
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



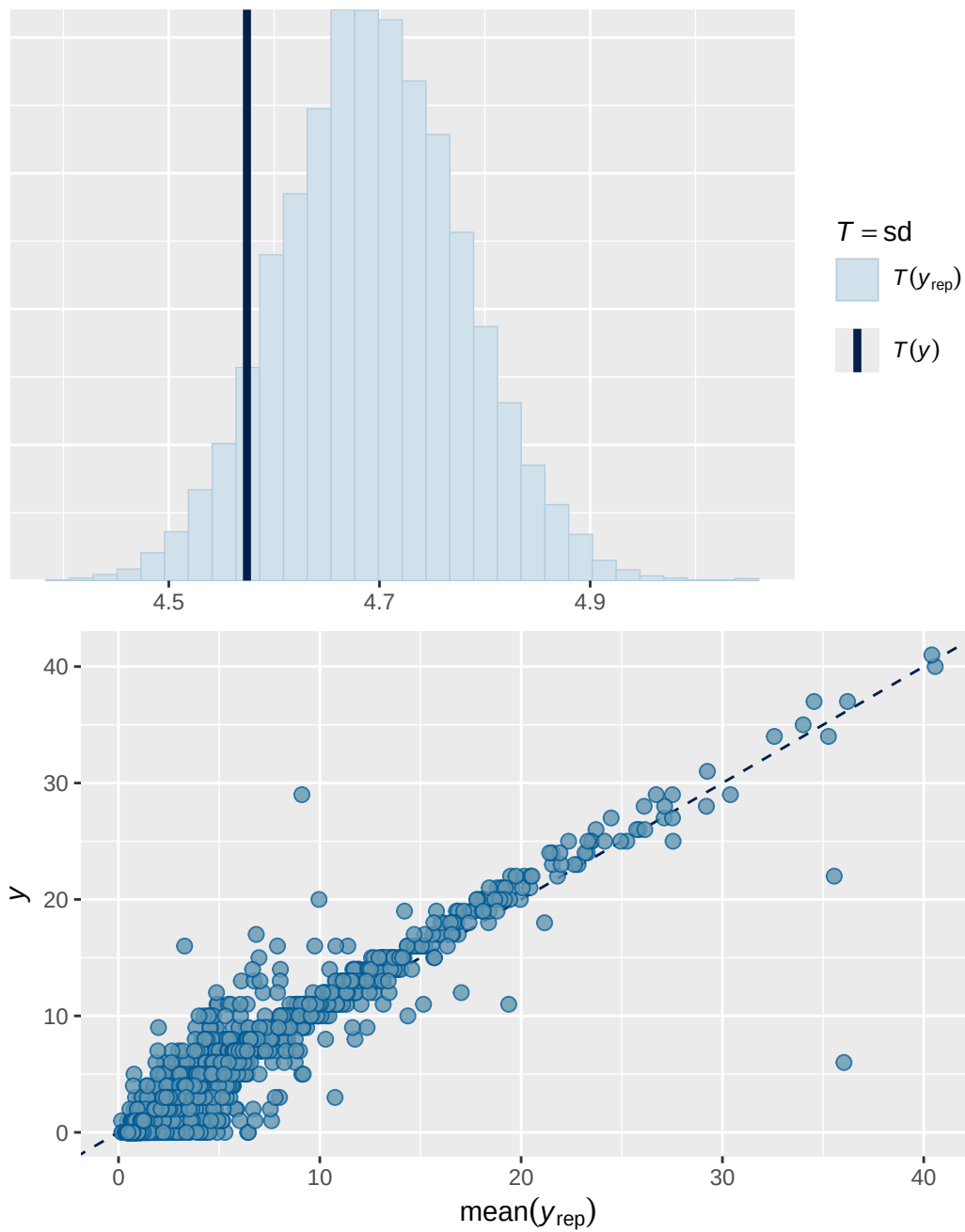
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 4 longitudinal output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.827	0.00457	0.818	0.836

Family: poisson
 Links: mu = log
 Formula: internalising_wave_3 ~ menarche_status_p_wave_2 * adhd_traits_wave_2_sc + age_wave_2_sc
 Data: analytic_df_wide (Number of observations: 4496)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.819	0.014	0.792	0.849	1.001	965	1671

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.109	0.030	0.059	0.177	1.001	704	1071

Regression Coefficients:

	Estimate	Est.Error	1-95% CI
Intercept	1.237	0.035	1.167
menarche_status_p_wave_2Y	0.174	0.031	0.115
adhd_traits_wave_2_sc	0.382	0.034	0.316
age_wave_2_sc	-0.021	0.030	-0.081
ethnraceblack_nh	-0.414	0.055	-0.526
ethnracehispanic	-0.091	0.047	-0.183
ethnraceother	-0.010	0.051	-0.112
inr_wave_0_sc	0.092	0.035	0.025
internalising_wave_0_sc	0.781	0.028	0.723
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	-0.036	0.050	-0.136

	u-95% CI	Rhat	Bulk_ESS
Intercept	1.308	1.000	1407
menarche_status_p_wave_2Y	0.234	1.001	1323
adhd_traits_wave_2_sc	0.447	1.000	995
age_wave_2_sc	0.037	1.000	1203
ethnraceblack_nh	-0.310	1.001	999
ethnracehispanic	0.001	1.002	983
ethnraceother	0.088	1.001	1198
inr_wave_0_sc	0.162	1.002	1098
internalising_wave_0_sc	0.832	1.000	982
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	0.061	1.002	1065

	Tail_ESS
Intercept	2018
menarche_status_p_wave_2Y	1898
adhd_traits_wave_2_sc	1832

age_wave_2_sc	2071
ethnraceblack_nh	1906
ethnracehispanic	1672
ethnraceother	1857
inr_wave_0_sc	1835
internalising_wave_0_sc	1614
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	1724

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 4 longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

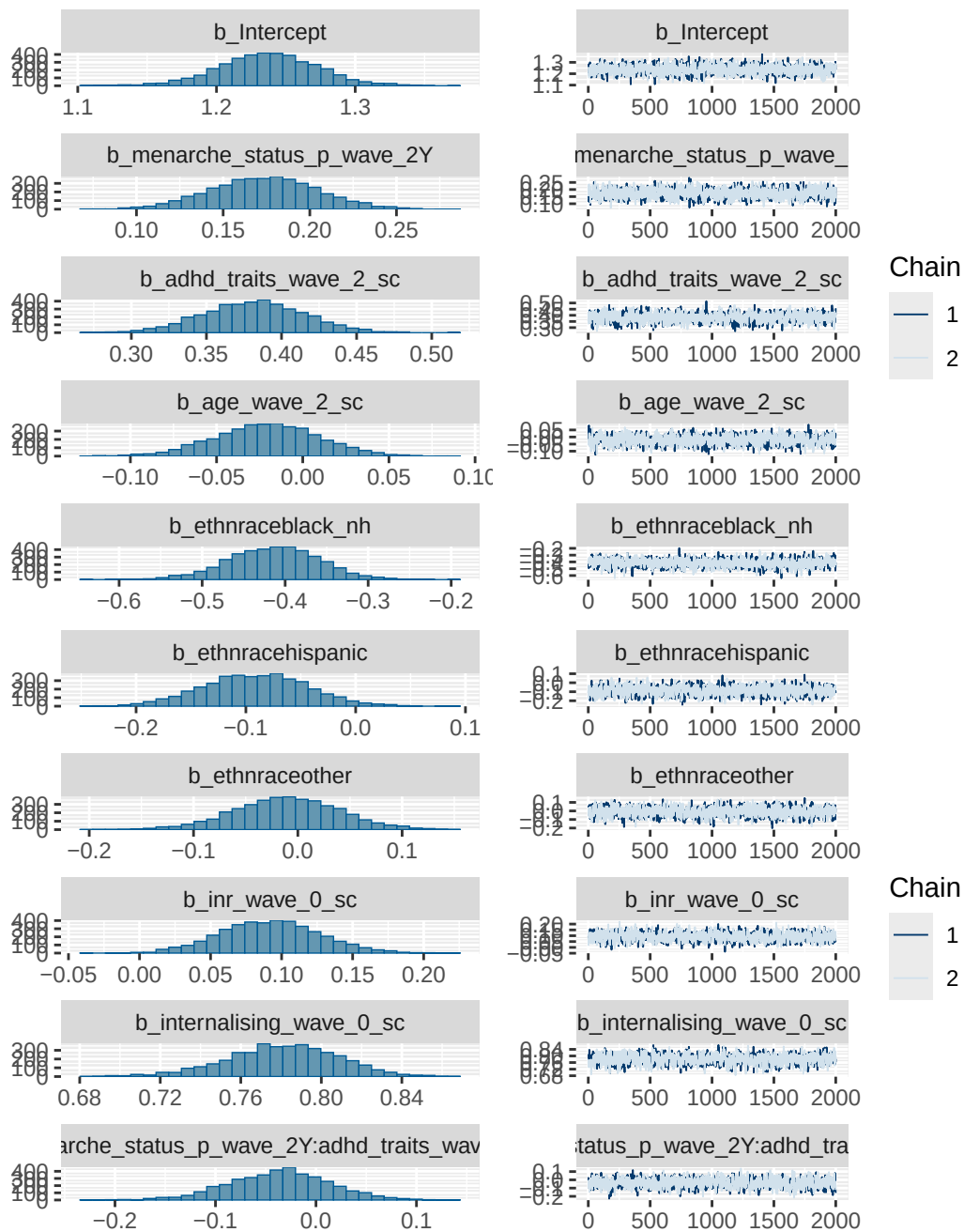
```
# A tibble: 3 x 4
```

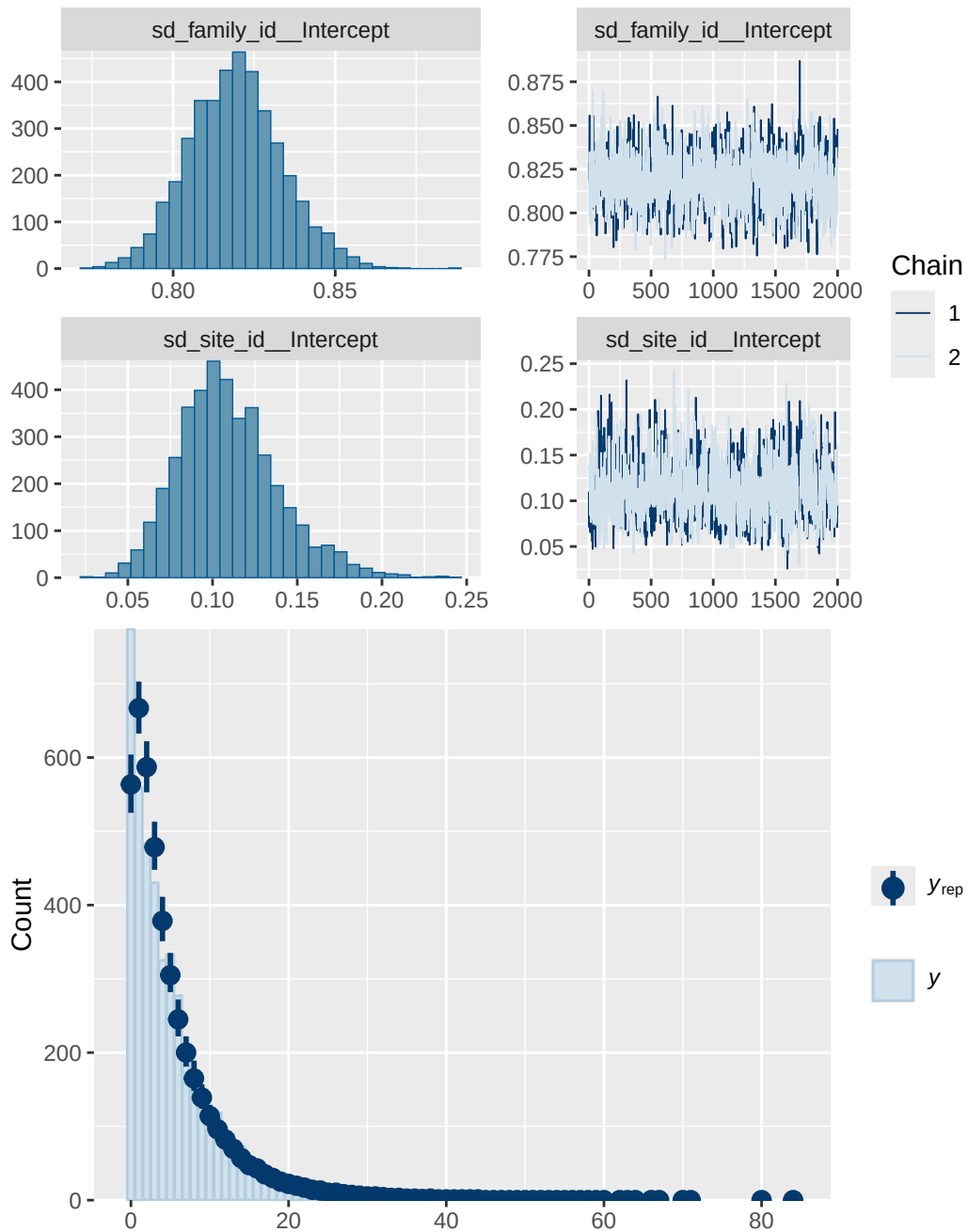
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche Y (Wave 2) [ADHD traits = mea~	19.0	12.1	26.4
2 ADHD traits (Wave 2) [Menarche = 0]	7.23	5.95	8.51
3 Menarche × ADHD traits	-0.66	-2.45	1.11

Model 4 longitudinal diagnostics

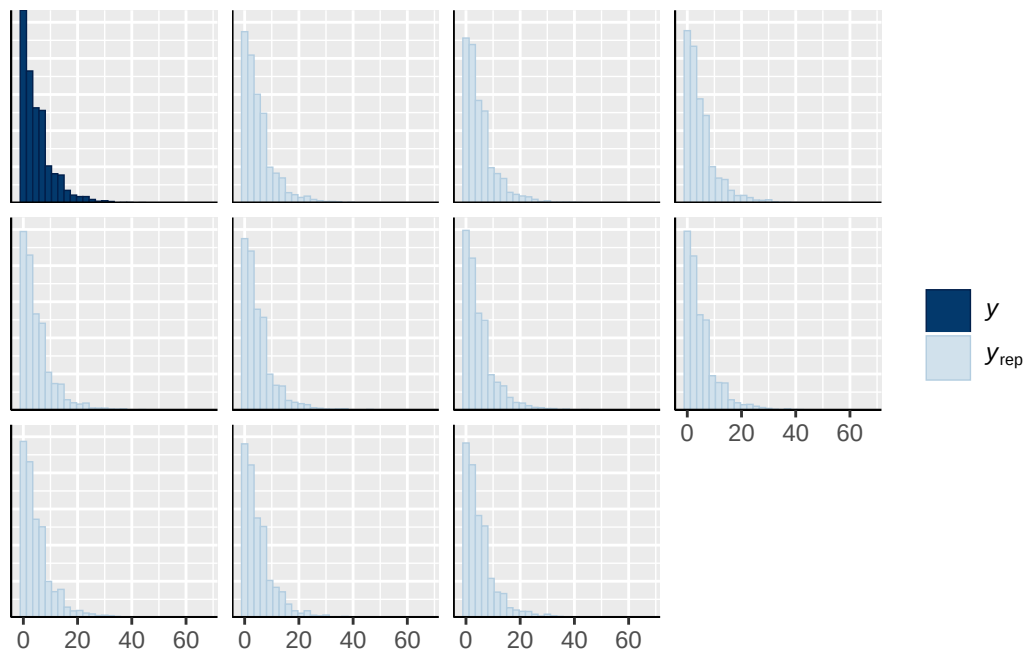
prior	class
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
normal(0, 1)	b
student_t(3, 1.4, 2.5) Intercept	
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd
student_t(3, 0, 2.5)	sd

		coef	group	resp	dpar	nlpar	lb
		adhd_traits_wave_2_sc					
		age_wave_2_sc					
		ethnraceblack_nh					
		ethnracehispanic					
		ethnraceother					
		inr_wave_0_sc					
		internalising_wave_0_sc					
		menarche_status_p_wave_2Y					
		menarche_status_p_wave_2Y:adhd_traits_wave_2_sc					
							0
			family_id				0
		Intercept	family_id				0
			site_id				0
		Intercept	site_id				0
ub tag	source						
	user						
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	default						
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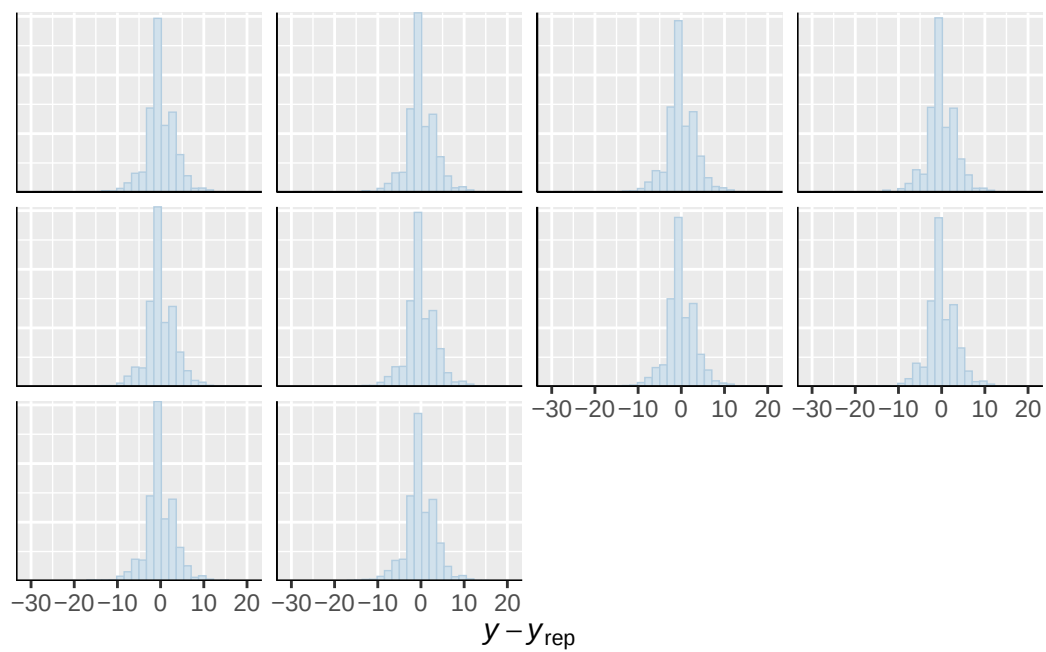




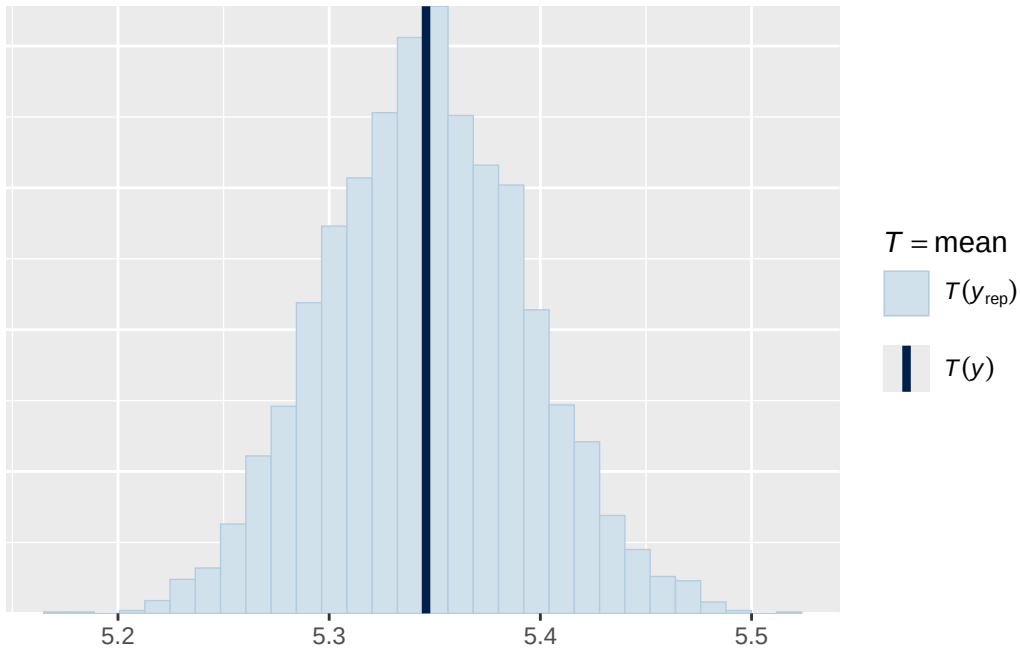
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



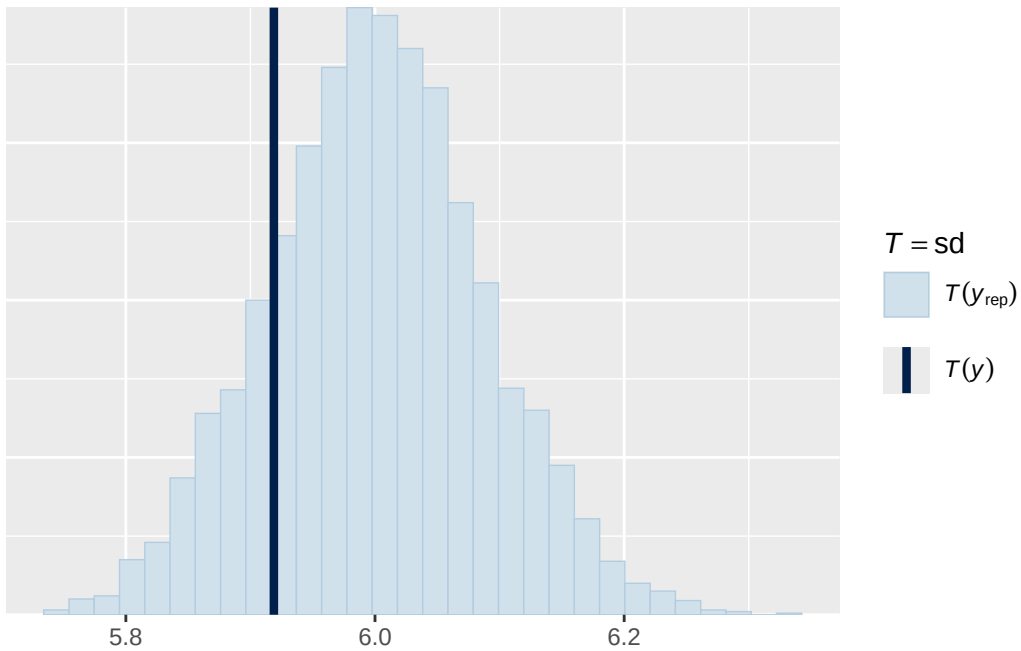
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

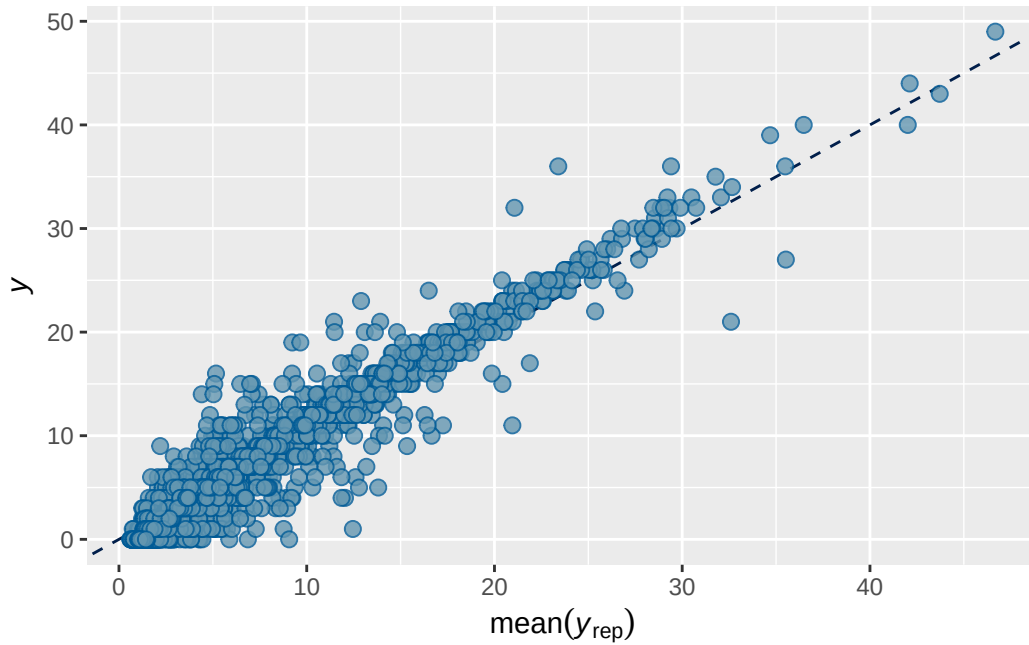


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Model 4b longitudinal output

```

      Estimate Est.Error  Q2.5 Q97.5
R2      0.828    0.00456 0.819 0.837

```

```

Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.824	0.015	0.796	0.853	1.003	980	2043

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.113	0.031	0.059	0.185	1.003	799	1251

Regression Coefficients:

	Estimate	Est.Error		
Intercept	1.308	0.035		
breast_development_p_wave_2_sc	0.157	0.029		
adhd_traits_wave_2_sc	0.365	0.027		
age_wave_2_sc	0.007	0.029		
ethnraceblack_nh	-0.409	0.055		
ethnracehispanic	-0.076	0.049		
ethnraceother	-0.022	0.052		
inr_wave_0_sc	0.073	0.033		
internalising_wave_0_sc	0.777	0.027		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.069	0.053		
	1-95% CI	u-95% CI	Rhat	
Intercept	1.238	1.376	1.001	
breast_development_p_wave_2_sc	0.098	0.212	1.001	
adhd_traits_wave_2_sc	0.312	0.419	1.005	
age_wave_2_sc	-0.051	0.063	1.002	
ethnraceblack_nh	-0.517	-0.300	1.003	
ethnracehispanic	-0.172	0.022	1.001	
ethnraceother	-0.125	0.074	1.001	
inr_wave_0_sc	0.008	0.138	1.001	
internalising_wave_0_sc	0.724	0.831	1.004	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	-0.032	0.175	1.000	
	Bulk_ESS	Tail_ESS		
Intercept	1264	1764		
breast_development_p_wave_2_sc	1361	2228		
adhd_traits_wave_2_sc	1043	1803		
age_wave_2_sc	1083	2146		
ethnraceblack_nh	934	1608		
ethnracehispanic	897	1754		
ethnraceother	922	1192		
inr_wave_0_sc	1172	2017		
internalising_wave_0_sc	860	1577		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	1106	2248		

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 4b longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

A tibble: 3 x 4

Variable <chr>	Estimate <dbl>	Lower 95% CI <dbl>	Upper 95% CI <dbl>
1 Breast development (Wave 2) [ADHD trai~	12.8	7.81	17.7
2 ADHD traits (Wave 2) [Breast developme~	6.9	5.88	7.96
3 Breast development × ADHD traits	0.97	-0.45	2.49

Model 4b longitudinal diagnostics

```

prior      class
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
student_t(3, 1.4, 2.5) Intercept
student_t(3, 0, 2.5)      sd
student_t(3, 0, 2.5)      sd
student_t(3, 0, 2.5)      sd
student_t(3, 0, 2.5)      sd
student_t(3, 0, 2.5)      sd
student_t(3, 0, 2.5)      sd

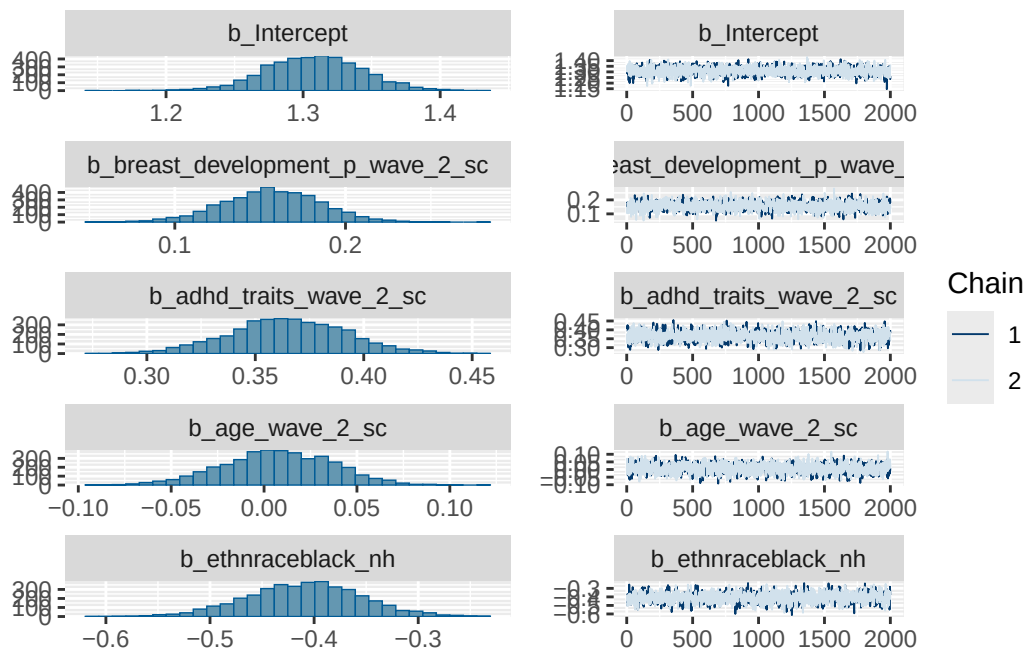
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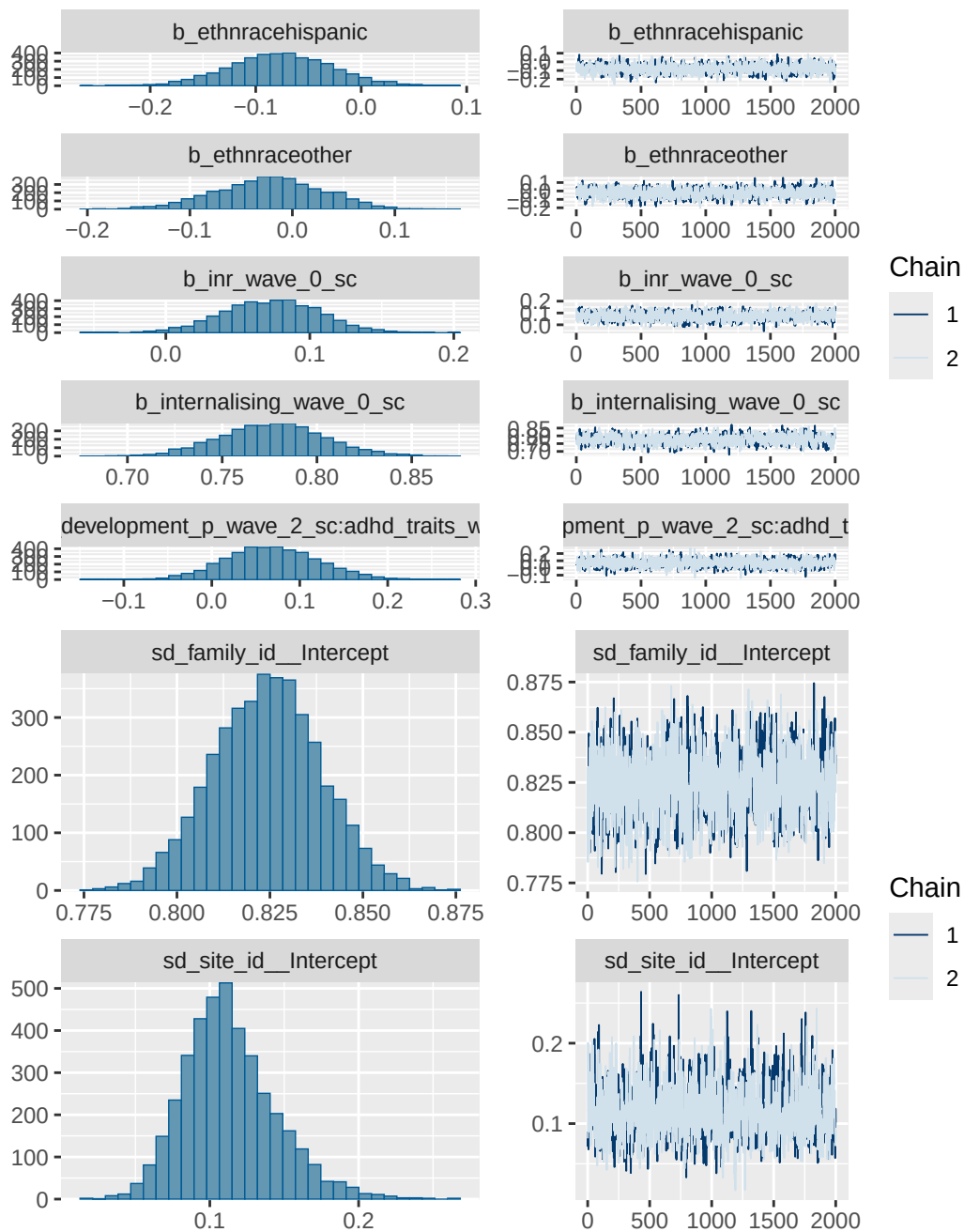
adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
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ethnracehispanic
ethnraceother
inr_wave_0_sc
internalising_wave_0_sc

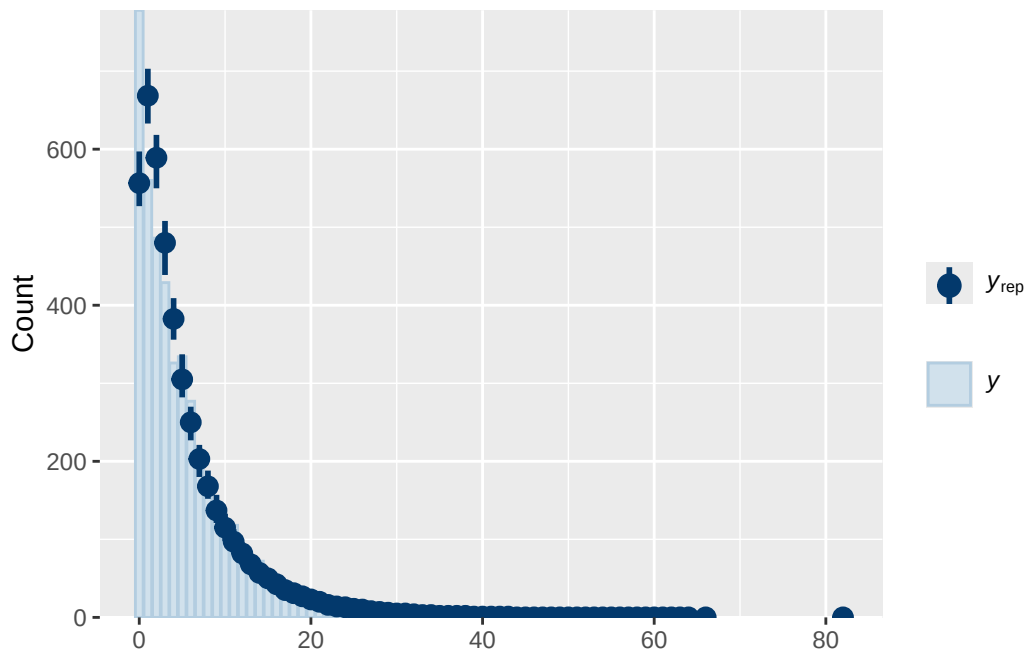
family_id
Intercept family_id
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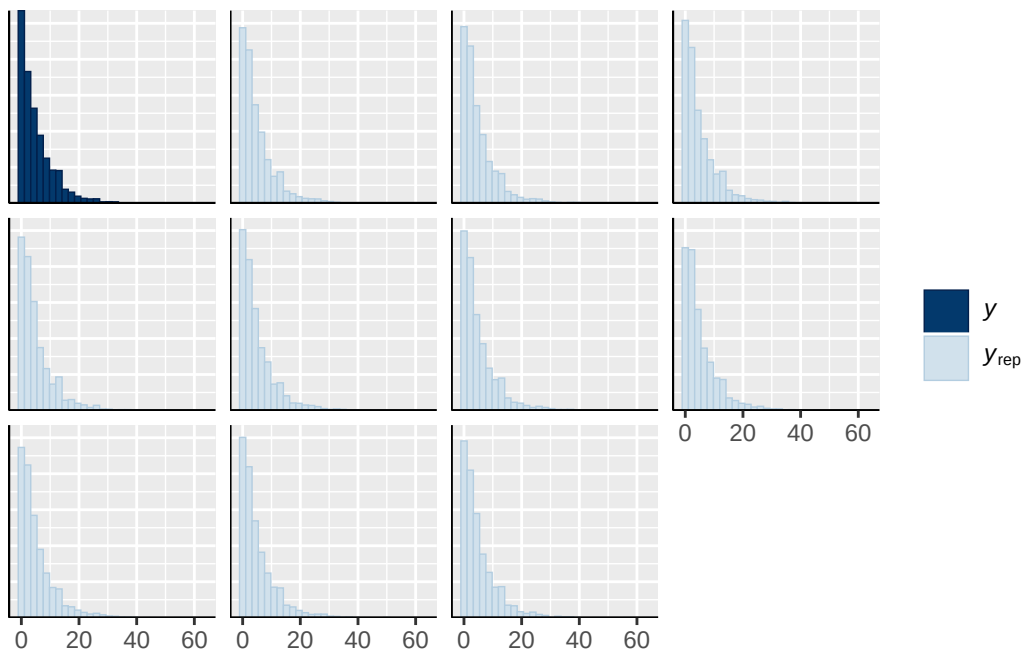
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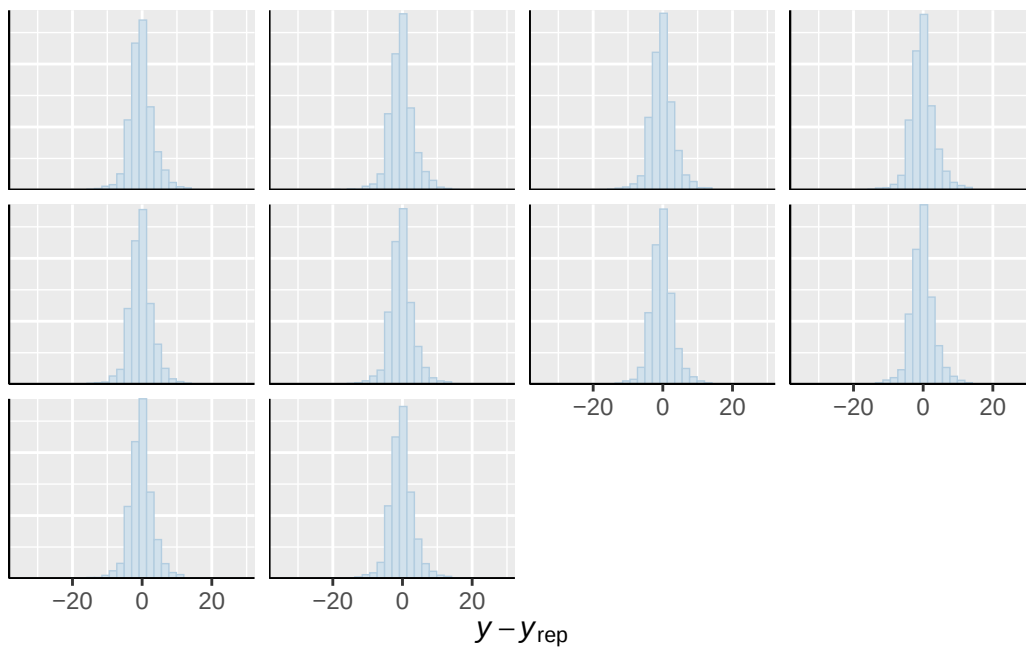




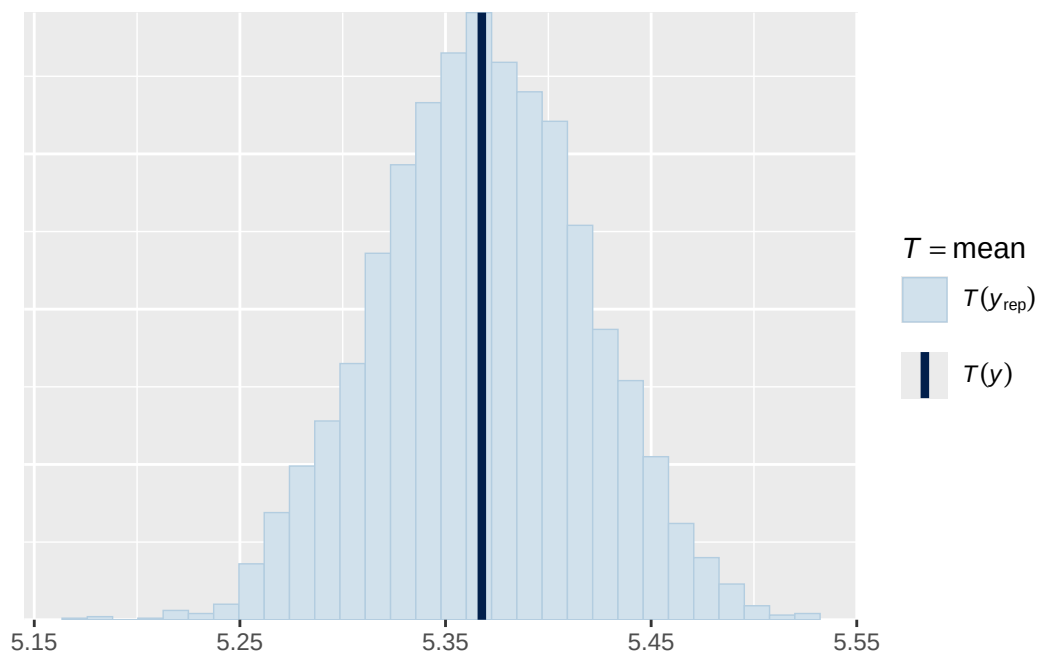
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



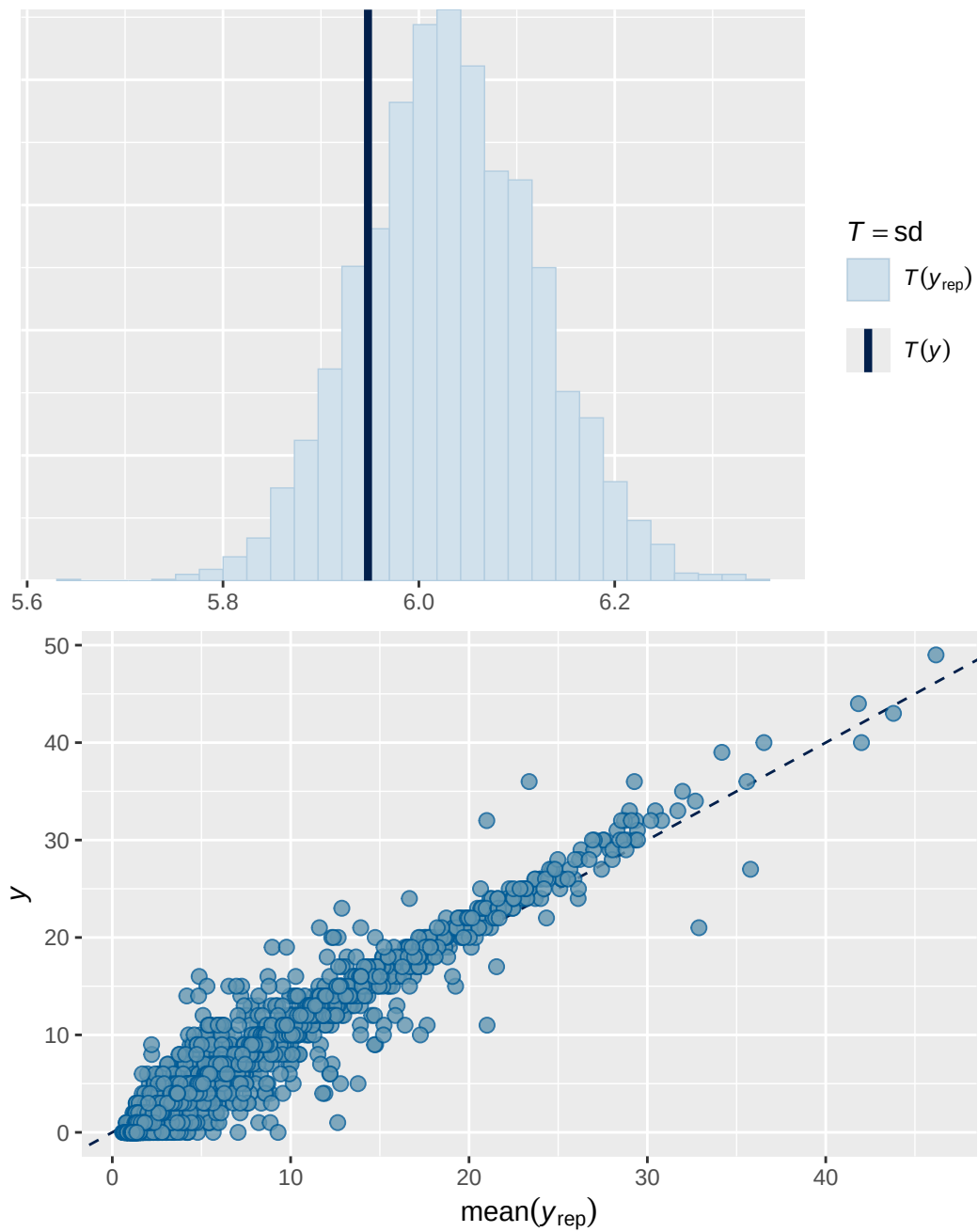
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 5 longitudinal output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.819	0.00541	0.808	0.829

Family: poisson
 Links: mu = log
 Formula: externalising_wave_3 ~ menarche_status_p_wave_2 * adhd_traits_wave_2_sc + age_wave_2_sc
 Data: analytic_df_wide (Number of observations: 4496)
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.976	0.020	0.938	1.015	1.000	2304	4106

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.071	0.038	0.007	0.150	1.005	648	1482

Regression Coefficients:

	Estimate	Est.Error	l-95% CI
Intercept	0.466	0.038	0.391
menarche_status_p_wave_2Y	0.112	0.041	0.033
adhd_traits_wave_2_sc	0.546	0.043	0.461
age_wave_2_sc	-0.104	0.038	-0.179
ethnraceblack_nh	-0.115	0.065	-0.238
ethnracehispanic	-0.103	0.059	-0.218
ethnraceother	-0.057	0.064	-0.183
inr_wave_0_sc	0.009	0.043	-0.078
externalising_wave_0_sc	0.914	0.034	0.847
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	0.092	0.062	-0.031

	u-95% CI	Rhat	Bulk_ESS
Intercept	0.541	1.000	3187
menarche_status_p_wave_2Y	0.191	1.001	3377
adhd_traits_wave_2_sc	0.629	1.001	2364
age_wave_2_sc	-0.030	1.001	3013
ethnraceblack_nh	0.014	1.000	2329
ethnracehispanic	0.014	1.000	2648
ethnraceother	0.068	1.000	2636
inr_wave_0_sc	0.095	1.000	2858
externalising_wave_0_sc	0.981	1.000	1760
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	0.213	1.001	2226

	Tail_ESS
Intercept	4380
menarche_status_p_wave_2Y	5376
adhd_traits_wave_2_sc	4010

age_wave_2_sc	3542
ethnraceblack_nh	3791
ethnracehispanic	4254
ethnraceother	4081
inr_wave_0_sc	4565
externalising_wave_0_sc	3019
menarche_status_p_wave_2Y:adhd_traits_wave_2_sc	3768

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5 longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

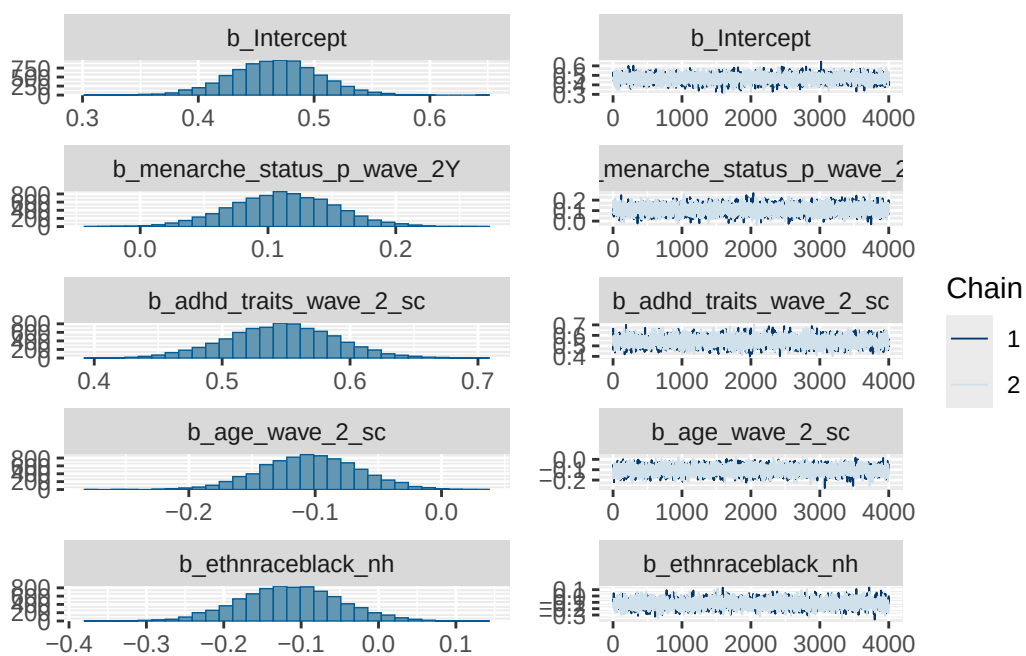
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# A tibble: 3 x 4
```

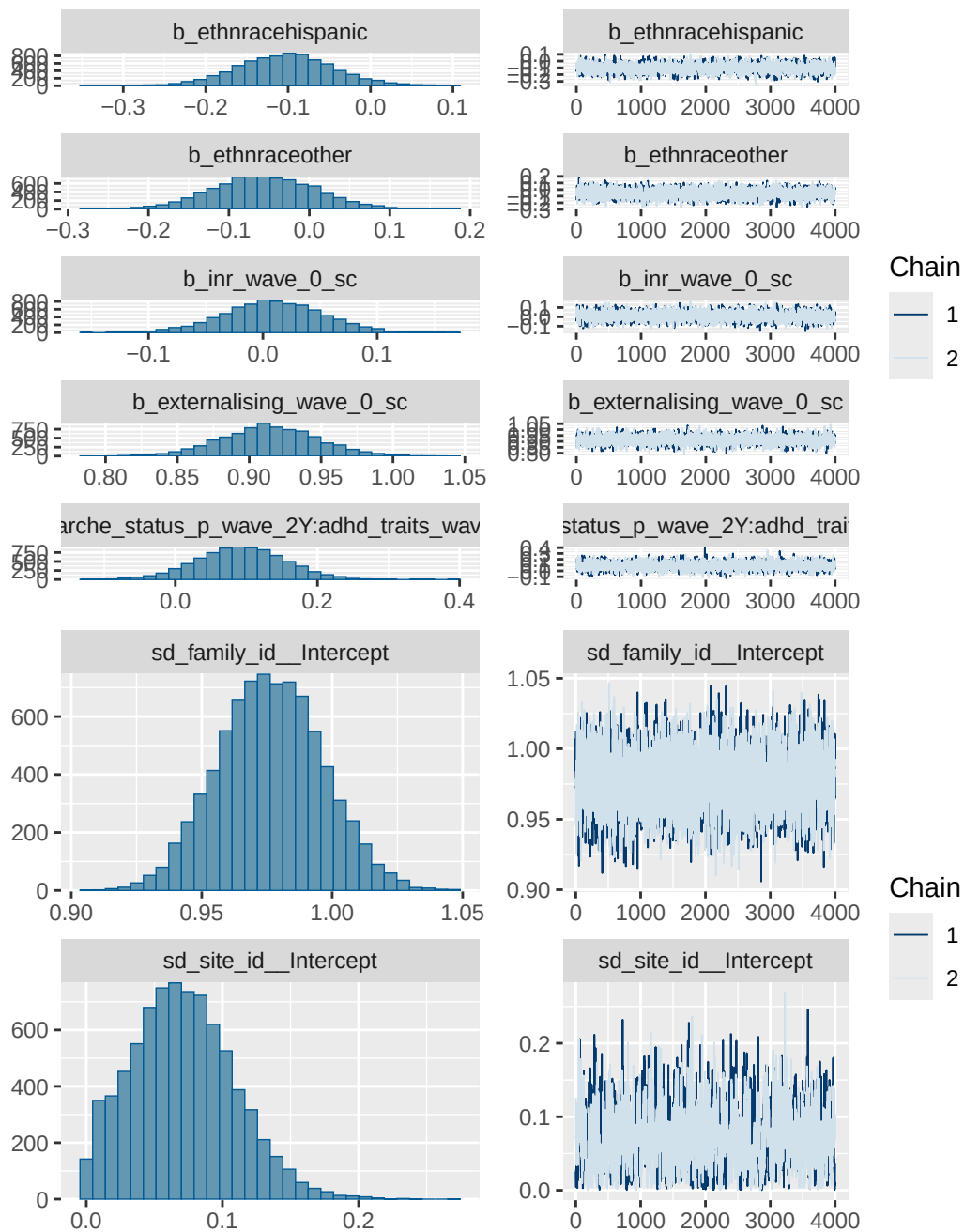
Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Menarche (Wave 2) [ADHD traits = mean]	11.8	3.31	21.1
2 ADHD traits (Wave 2) [Menarche = 0]	10.5	8.79	12.2
3 Menarche × ADHD traits	1.7	-0.56	3.97

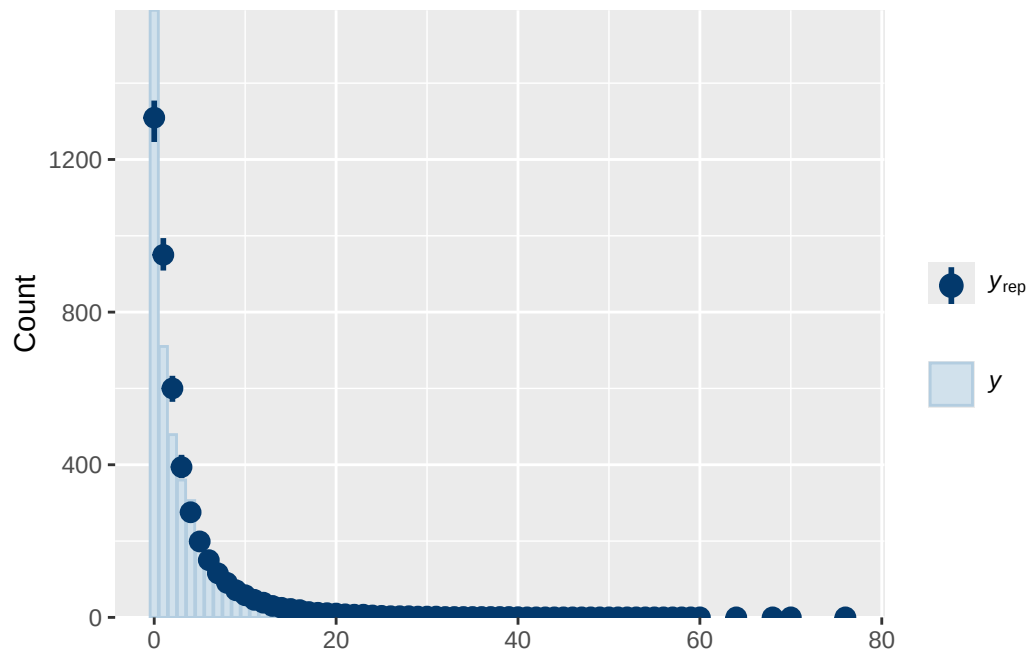
Model 5 longitudinal diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_traits_wave_2_sc
	normal(0, 1)	b	age_wave_2_sc
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	externalising_wave_0_sc
	normal(0, 1)	b	inr_wave_0_sc
	normal(0, 1)	b	menarche_status_p_wave_2Y
	normal(0, 1)	b	menarche_status_p_wave_2Y:adhd_traits_wave_2_sc
student_t(3, 0, 2.9)		Intercept	
student_t(3, 0, 2.9)		sd	
student_t(3, 0, 2.9)		sd	
student_t(3, 0, 2.9)		sd	Intercept
student_t(3, 0, 2.9)		sd	
student_t(3, 0, 2.9)		sd	Intercept

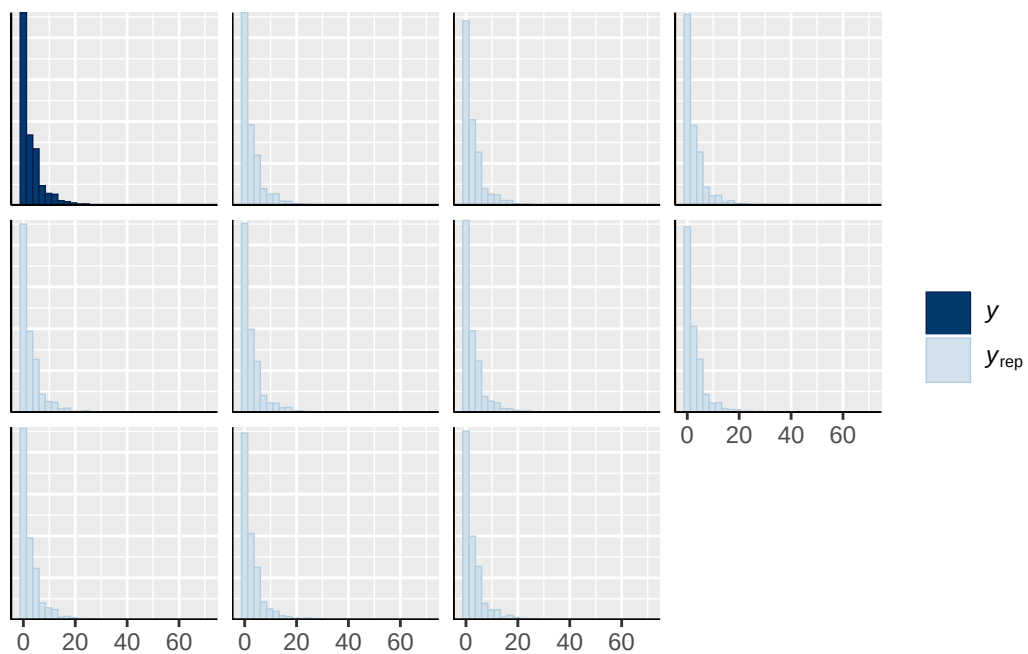
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site_id				0			(vectorized)
site_id				0			(vectorized)



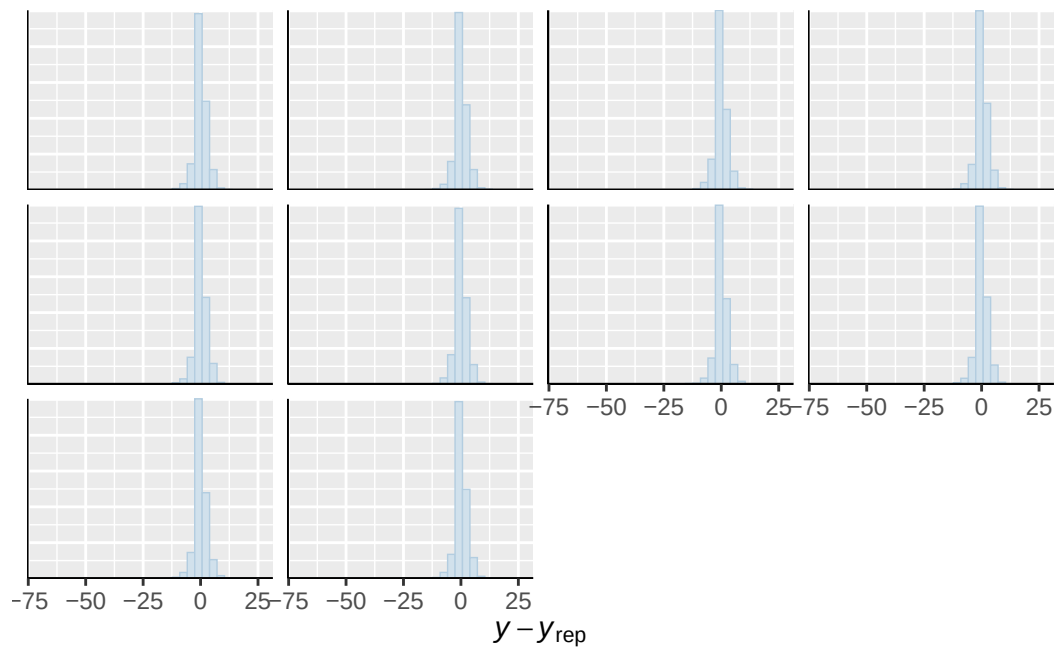




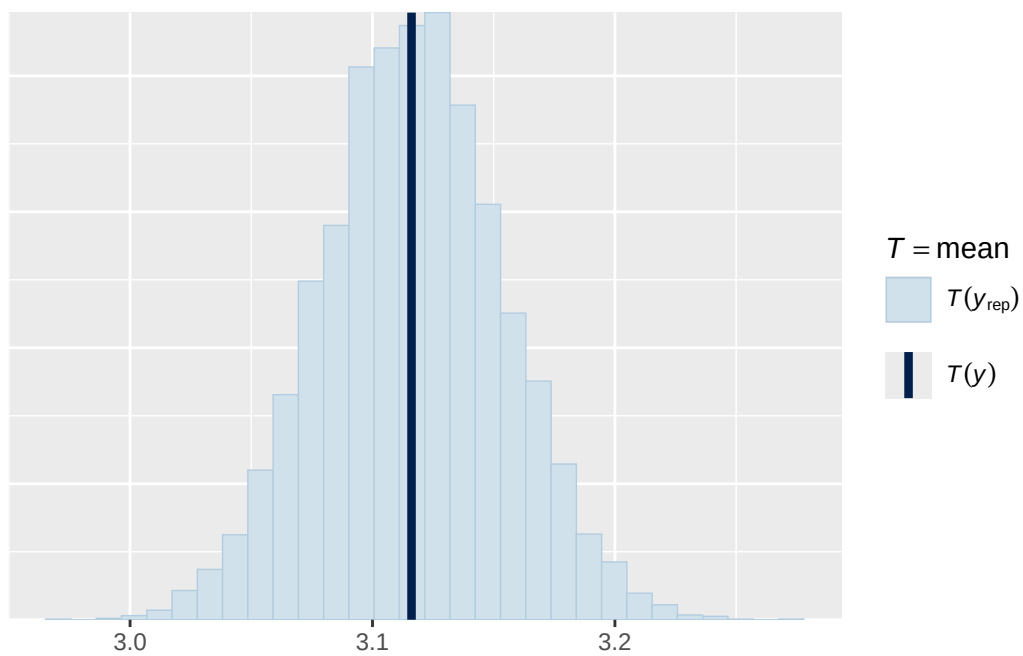
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



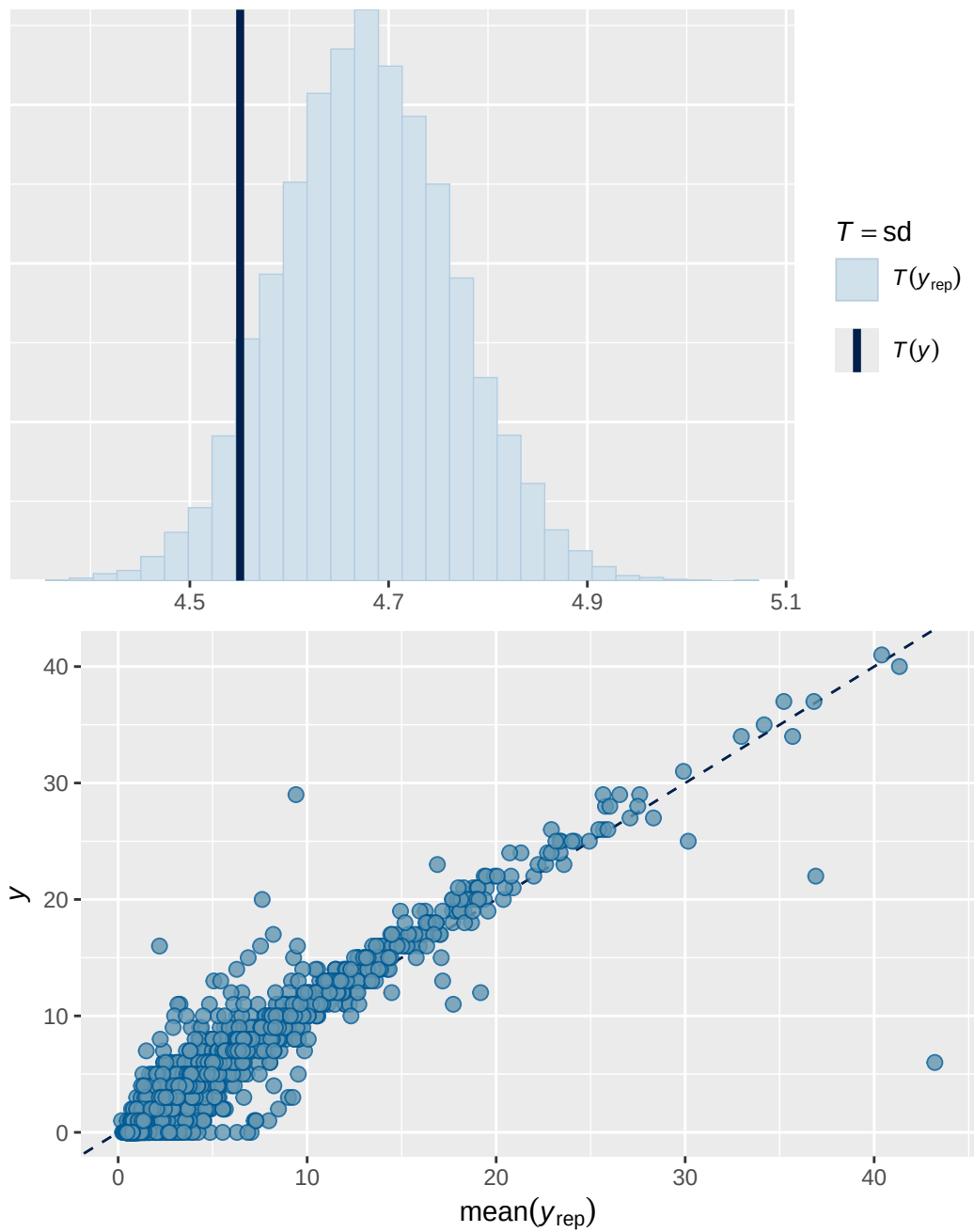
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Model 5b longitudinal output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.823	0.00521	0.813	0.833

Family: poisson
 Links: mu = log
 Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age.
 Data: analytic_df_wide (Number of observations: 4515)
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.985	0.020	0.948	1.026	1.002	2022	3797

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.065	0.037	0.005	0.143	1.010	487	1393

Regression Coefficients:

	Estimate	Est.Error
Intercept	0.510	0.034
breast_development_p_wave_2_sc	0.189	0.039
adhd_traits_wave_2_sc	0.586	0.034
age_wave_2_sc	-0.117	0.037
ethnraceblack_nh	-0.131	0.064
ethnracehispanic	-0.098	0.056
ethnraceother	-0.079	0.064
inr_wave_0_sc	-0.002	0.044
externalising_wave_0_sc	0.913	0.035
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.174	0.065

	1-95% CI	u-95% CI	Rhat
Intercept	0.442	0.576	1.000
breast_development_p_wave_2_sc	0.112	0.264	1.001
adhd_traits_wave_2_sc	0.516	0.652	1.002
age_wave_2_sc	-0.190	-0.046	1.002
ethnraceblack_nh	-0.257	-0.003	1.001
ethnracehispanic	-0.206	0.012	1.000
ethnraceother	-0.206	0.043	1.001
inr_wave_0_sc	-0.088	0.084	1.003
externalising_wave_0_sc	0.845	0.983	1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.041	0.299	1.000

	Bulk_ESS	Tail_ESS
Intercept	2612	4465
breast_development_p_wave_2_sc	2861	4312
adhd_traits_wave_2_sc	1899	3563

age_wave_2_sc	2452	3920
ethnraceblack_nh	1956	3539
ethnracehispanic	1980	3800
ethnraceother	1858	3008
inr_wave_0_sc	1692	3173
externalising_wave_0_sc	1640	3223
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	1916	2940

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5b longitudinal effects expressed as % change in expected outcome per 1-unit increase in original predictor scale

```
# A tibble: 3 x 4
```

Variable <chr>	Estimate <dbl>	`Lower 95% CI` <dbl>	`Upper 95% CI` <dbl>
1 Breast development (Wave 2) [ADHD trai~	15.6	8.99	22.5
2 ADHD traits (Wave 2) [Breast developme~	11.3	9.9	12.7
3 Breast development × ADHD traits	2.47	0.57	4.29

Model 5b interaction analysis

	Variable	Estimate	Lower 95% CI
1	ADHD traits (Wave 2) → Outcome [Breast development = 1]	6.92	3.58
2	ADHD traits (Wave 2) → Outcome [Breast development = 4]	15.06	11.93
3	Difference in slope (Stage 4 vs Stage 1)	-7.06	-

12.23

	Upper 95% CI
1	10.50
2	18.17
3	-2.16

Model 5b interaction plot

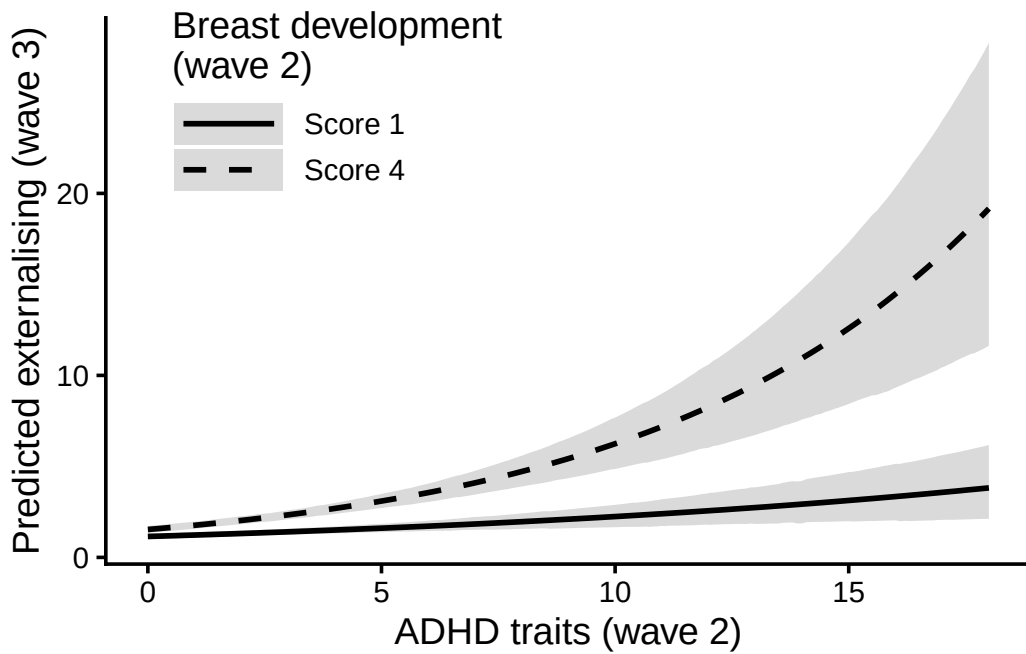
ADHD mean: 2

ADHD SD: 2.73

ADHD range: 0 18

	adhd_traits_wave_2_sc	breast_development_p_wave_2_sc	rate	lower.HPD	upper.HPD
1	-0.366	-1.26	1.15	0.996	1.31
2	-0.347	-1.26	1.16	1.007	1.32
3	-0.329	-1.26	1.16	1.013	1.32
4	-0.311	-1.26	1.17	1.021	1.33
5	-0.293	-1.26	1.18	1.031	1.33
6	-0.274	-1.26	1.19	1.041	1.34

	adhd_raw	breast_dev
1	0.0	Score 1
2	0.1	Score 1
3	0.2	Score 1
4	0.3	Score 1
5	0.4	Score 1
6	0.5	Score 1



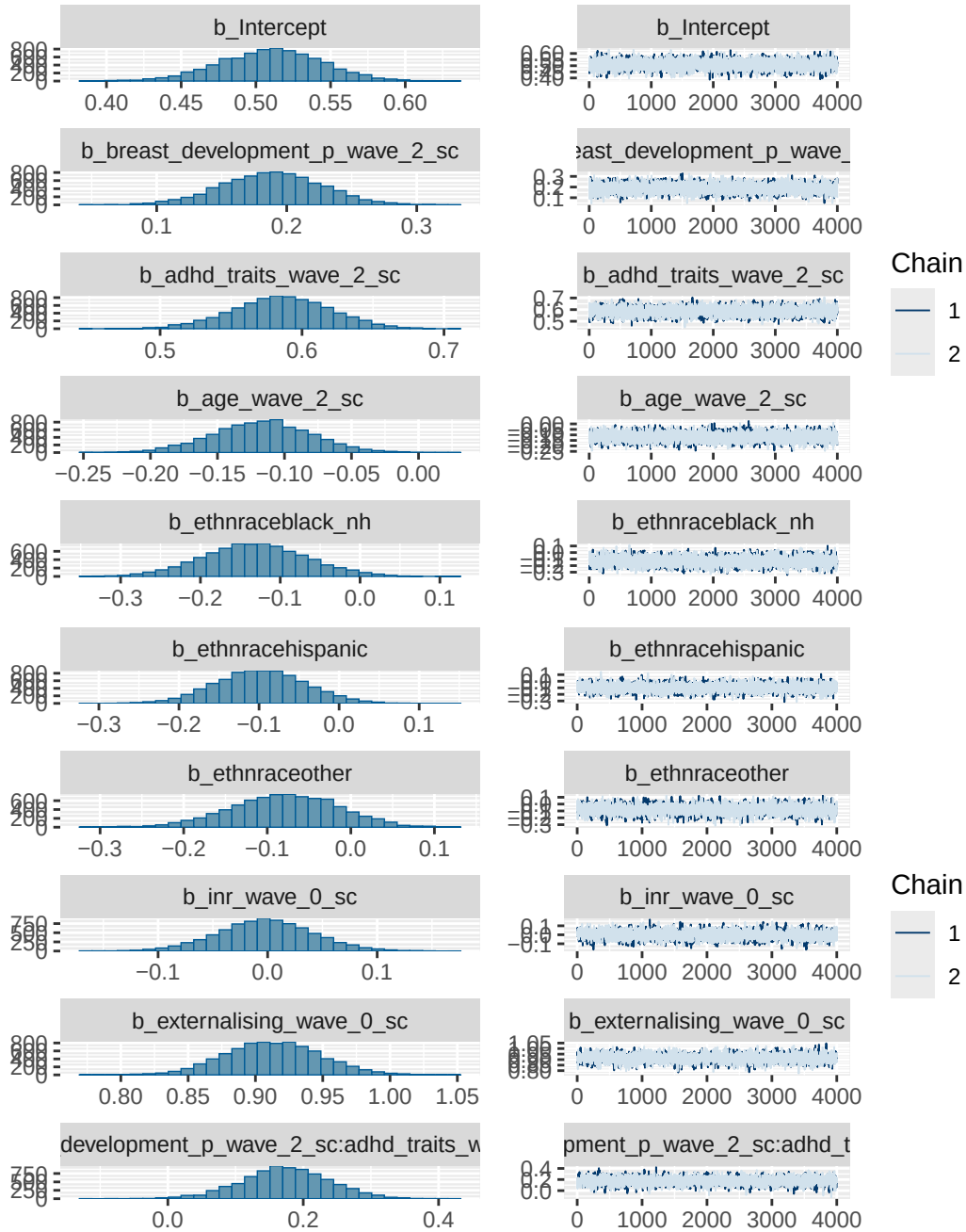
Model 5b longitudinal diagnostics

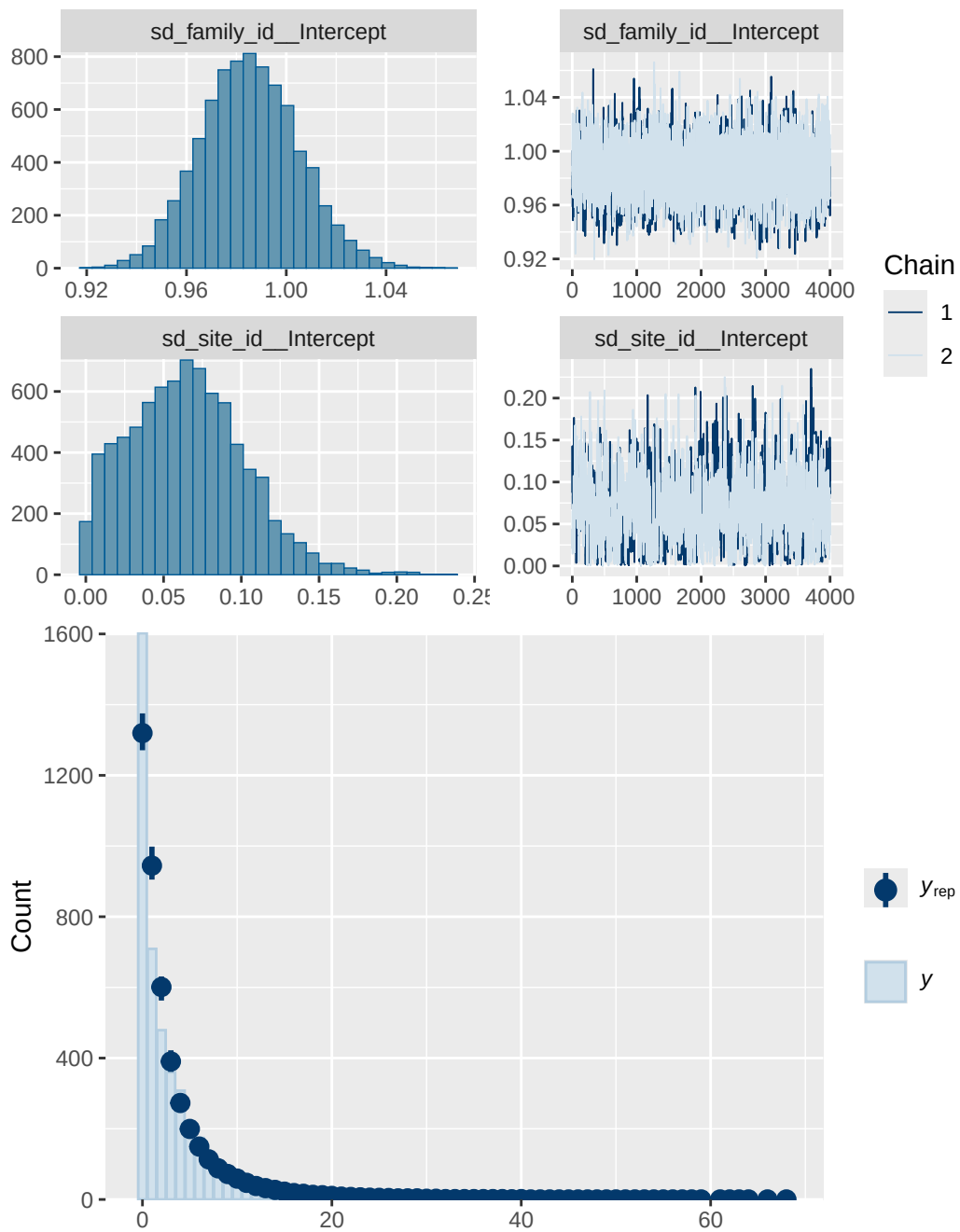
prior	class
normal(0, 1)	b
normal(0, 1)	b


```

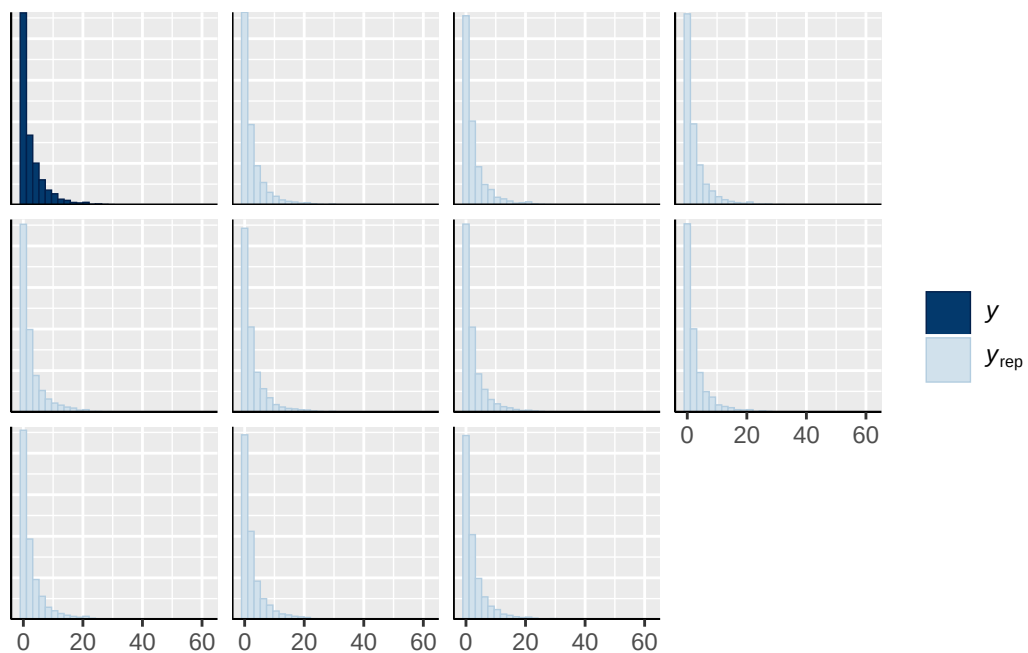
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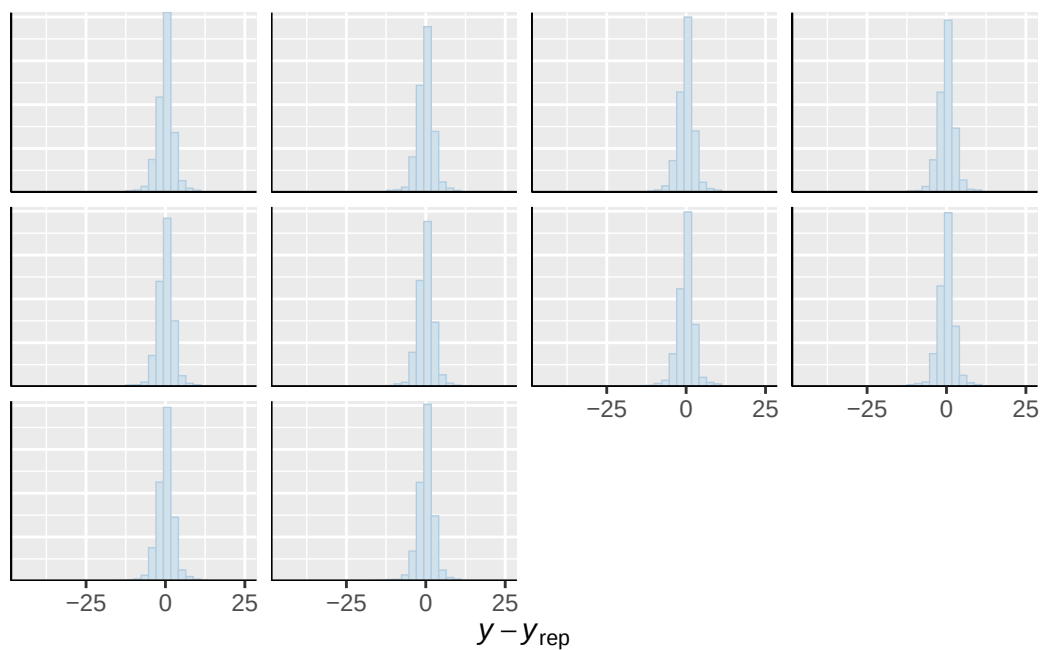




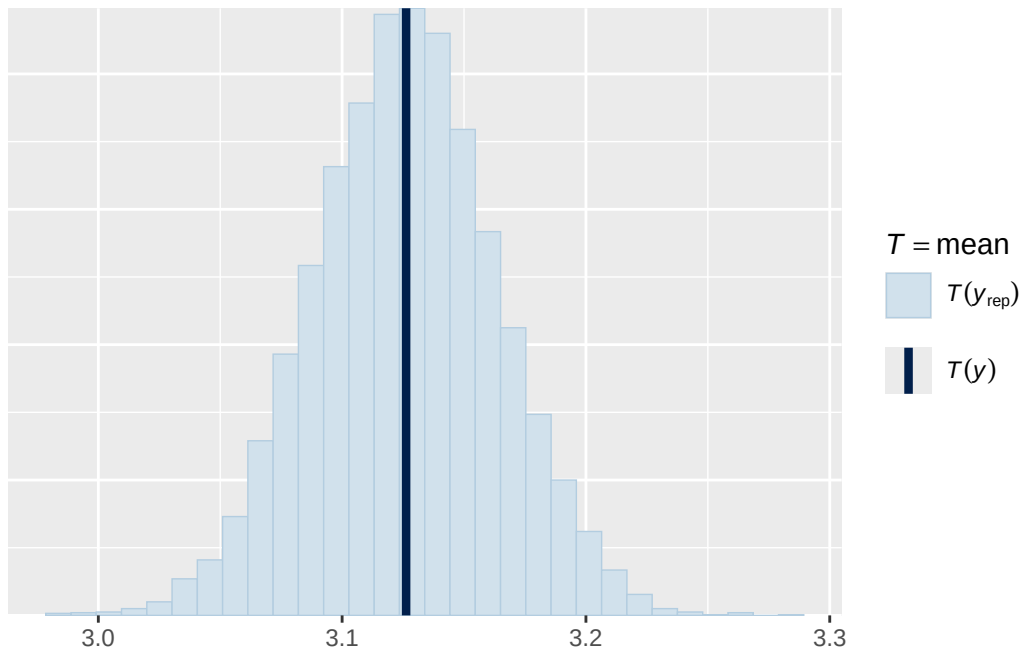
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

